COHERENT QUADRANT

Battery Fire Safety Industry Analysis

India

97 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

Battery Fire Safety Industry Analysis

Market Type

The market encompasses technologies, materials, and systems designed to prevent, detect, contain, and suppress fires and thermal runaway events in lithium-ion and other advanced battery systems.

Provider Categories in this Market

1. Manufacturer

Manufacturer was chosen because the market's defining competitive landscape is built around companies producing and selling physical containment materials, barriers, and specialized suppression hardware that physically prevent or mitigate thermal runaway.

97	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

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Market Scoring Parameters

X — Product Capability

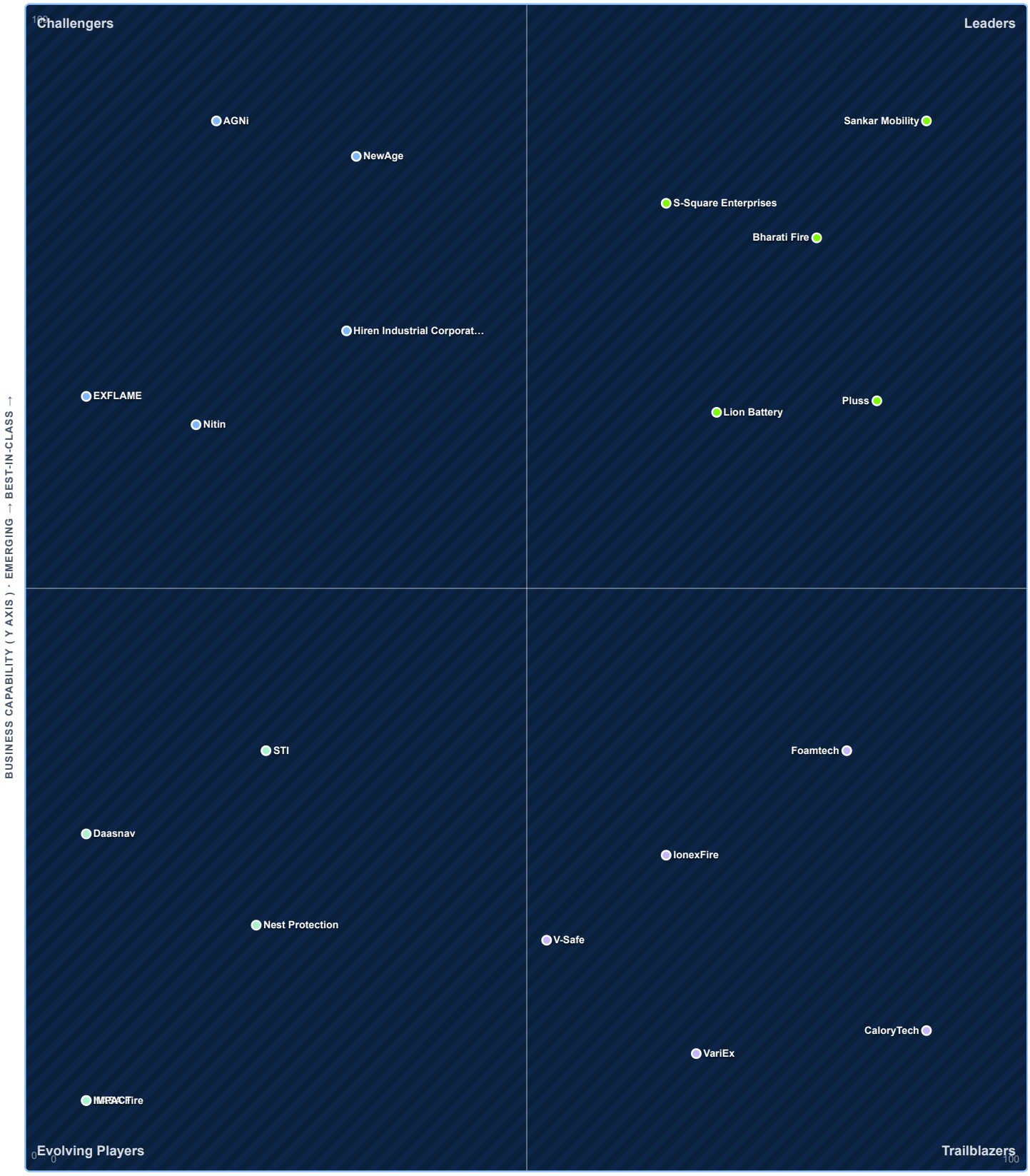
1. **Fire-Resistant Battery Chemistry & Materials**
2. **Thermal Runaway Prevention & Suppression Systems**
3. **Battery Management System (BMS) Safety Features**
4. **Compliance with Indian & International Safety Standards**
5. **R&D and Innovation in Battery Fire Safety**

Y — Business Capability

1. **Tier-1 OEM Partnerships & Supply Contracts**
2. **Revenue & Growth in Indian Market**
3. **Manufacturing Capacity & Localization in India**
4. **After-Sales Service & Support Network**
5. **Regulatory Approvals & Certifications**

Quadrant Positioning

(20 on chart · 97 in total)



Countries on chart (by quadrant)

Leaders: India×5 · **Challengers**: India×5 · **Trailblazers**: India×5 · **Evolving Players**: India×5

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Sankar Mobility	Sankar Advanced Mobility Solutions Pvt. Ltd.	Chennai, India	Manufacturer	Leaders	99	93	96
Bharati Fire	Bharati Fire Engineers	Mumbai, India	Manufacturer	Leaders	87	92	90
Pluss	Pluss Advanced Technologies Pvt. Ltd. (subsidiary of Carborundum Universal Limited)	Gurugram, India	Manufacturer	Leaders	89	87	88
S-Square Enterprises	S-Square Enterprises	Pune, India	Manufacturer	Leaders	85	89	87
Lion Battery	Lion Battery	Valsad, India	Manufacturer	Leaders	85	87	86
AGNi	AGNI Devices Pvt. Ltd.	New Delhi, India	Manufacturer	Challengers	72	88	80
NewAge	NewAge Fire Protection Industries Pvt. Ltd.	Mumbai, India	Manufacturer	Challengers	72	87	80
Hiren Industrial Corporation	Hiren Industrial Corporation	Mumbai, India	Manufacturer	Challengers	78	80	79
Nitin	Nitin Fire Protection Industries Ltd.	Mumbai, India	Manufacturer	Challengers	71	78	74
EXFLAME	Exflame	New Delhi, India	Manufacturer	Challengers	40	77	58
Foamtech	Foamtech Antifire Company	Delhi, India	Manufacturer	Trailblazers	86	83	84
IonexFire	Ionex Fire	Gurgaon, India	Manufacturer	Trailblazers	83	80	82
CaloryTech	CaloryTech	Bangalore, India	Manufacturer	Trailblazers	87	78	82
VariEx	Varistor Technologies Pvt. Ltd.	Bengaluru, India	Manufacturer	Trailblazers	79	77	78
V-Safe	Varia FireTech Industries	Jamnagar, India	Manufacturer	Trailblazers	53	77	65
STI	Sheetla Techno Industries	Gurugram, India	Manufacturer	Evolving Players	76	74	75
Nest Protection	Nest Protection	Gurugram, India	Manufacturer	Evolving Players	75	73	74
Daasnav	Daasnav Solutions Private Limited	Noida, India	Manufacturer	Evolving Players	54	85	70
NASA Fire	Nasa Fire Technology Limited	Ahmedabad, India	Manufacturer	Evolving Players	60	64	62
IMPACT	Impact Fire & Safety Appliances Pvt. Ltd.	New Delhi, India	Manufacturer	Evolving Players	58	47	52

Other Noticeable Players

(77 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
Rubymica	Ruby Mica Company Limited	Giridih, India	Manufacturer	Leaders	
BLAZEFUZE	Jetik Innovation Private Limited	Delhi, India	Manufacturer	Leaders	
Fire Knight	Fire Knight	New Delhi, India	Manufacturer	Leaders	
Resguardo	Resguardo Industries Pvt. Ltd.	Bangalore, India	Manufacturer	Leaders	
SNF	Shimnit New Fire Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Ceasefire	Ceasefire Industries Private Limited	Noida, India	Manufacturer	Leaders	
Lion-Ex	United Fire Equipment Pvt. Ltd. (subsidiary of Great Sales Corporation)	New Delhi, India	Manufacturer	Leaders	
ARPL	Asean Rubber Pvt. Ltd.	Gurugram, India	Manufacturer	Leaders	
Kanex	Kanex Fire	Mumbai, India	Manufacturer	Leaders	
Murugappa Morgan	Murugappa Morgan Thermal Ceramics Ltd.	Chennai, India	Manufacturer	Leaders	
Firetech	Firetech Equipment & Systems Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Intime	Intime Fire Appliances Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Usha Fire	Usha Fire Safety Equipments Pvt. Ltd.	Chennai, India	Manufacturer	Leaders	
Adhesive Specialities	Adhesive Specialities	Bangalore, India	Manufacturer	Leaders	
Alkraft	Alkraft Thermotechnologies Pvt. Ltd.	Chennai, India	Manufacturer	Leaders	
Prosafe	Prosafe Safety Equipment's Pvt Ltd	New Delhi, India	Manufacturer	Leaders	
Wedge India	Wedge Industries Limited	New Delhi, India	Manufacturer	Leaders	
RNG Performance Materials	RNG Performance Materials	Vadodara, India	Manufacturer	Leaders	
Fire Engineering Technology	Fire Engineering Technology	New Delhi, India	Manufacturer	Leaders	
IFP UNILIGHT	Integrated Fire Protection Pvt. Ltd.	Kolkata, India	Manufacturer	Leaders	
MREX-Li6	KV Fire Chemicals (India) Pvt. Ltd.	Navi Mumbai, India	Manufacturer	Leaders	
WEFIRE	Warrier Electronics	Pune, India	Manufacturer	Leaders	
Swastik Synergy	Swastik Synergy Engineering Private Limited	Mumbai, India	Manufacturer	Leaders	
DARQ	Darq Industries Private Limited	Gurugram, India	Manufacturer	Leaders	
Modi International	Modi International	New Delhi, India	Manufacturer	Leaders	
Supremex	Supremex Equipments	Mumbai, India	Manufacturer	Leaders	
Safex	Safex Fire Services Limited	Mumbai, India	Manufacturer	Leaders	
NEXGEN	DSS Industries	Pune, India	Manufacturer	Leaders	
Signature	Darshan Safety Zone	Ahmedabad, India	Manufacturer	Leaders	
HEELIX	Heelix Protech LLP	Mumbai, India	Manufacturer	Leaders	
Palex	Palladium Safety Solutions Pvt. Ltd.	Gurgaon, India	Manufacturer	Leaders	
Supertech Fabrics	Supertech Fabrics Pvt. Ltd.	Vadodara, India	Manufacturer	Leaders	
TCS	Tek Chand & Sons	New Delhi, India	Manufacturer	Leaders	
Sheffer	Sheffer Industries	Mumbai, India	Manufacturer	Leaders	

ASHA	Asha Enterprise	Ahmedabad, India	Manufacturer	Trailblazers	
Axis Fire Protection	Axis Fire Protection	Gurugram, India	Manufacturer	Leaders	
Padmini	Padmini Industries Ltd.	Noida, India	Manufacturer	Challengers	
Vimal Fire	Vimal Fire Controls Pvt. Ltd.	Vadodara, India	Manufacturer	Leaders	
USAFE	United Fire and Safety Services	Vadodara, India	Manufacturer	Leaders	
FireChem	Fire Safety Devices Pvt. Ltd.	Faridabad, India	Manufacturer	Trailblazers	
FirePanda	Firepanda Private Limited	Bangalore, India	Manufacturer	Trailblazers	
Sea Max Fire	Sea Max Fire Engineering Works	New Delhi, India	Manufacturer	Challengers	
Fire Avengers	AR Fire Safety Solutions	New Delhi, India	Manufacturer	Trailblazers	
HD Fire Protect	HD Fire Protect Limited	Thane, India	Manufacturer	Challengers	
MITRAS	Mitras Technocrafts Private Limited	New Delhi, India	Manufacturer	Challengers	
Triangle Fire	Triangle Fire Safety India Private Limited	Pune, India	Manufacturer	Challengers	
EV FirePro	ASKA Equipments Private Limited	New Delhi, India	Manufacturer	Challengers	
balawa	Agni Controls	Chennai, India	Manufacturer	Trailblazers	
Ardent	Ardent Fire Systems	Gurugram, India	Manufacturer	Trailblazers	
SRS Enterprises	SRS Enterprises	Coimbatore, India	Manufacturer	Trailblazers	
ACC Insulations	Arunoday Composit Corporation	Nashik, India	Manufacturer	Trailblazers	
Sigma Fire	Sigma Fire Industries	Thane, India	Manufacturer	Trailblazers	
Technomatic	Technomatic Industries	Faridabad, India	Manufacturer	Challengers	
HiCare	Hicare Safety Solutions	Navi Mumbai, India	Manufacturer	Trailblazers	
FIRESPRAY	Drownfire Private Limited	Indore, India	Manufacturer	Evolving Players	
AFS	Advanced Fire & Safety India Pvt. Ltd.	Noida, India	Manufacturer	Evolving Players	
Daman Fire	Daman Fire Service	Ludhiana, India	Manufacturer	Evolving Players	
Endeavour	Endeavour Safety Solutions Private Limited	Gurugram, India	Manufacturer	Evolving Players	
Fire Shield	Fireshield Engineers (India) Limited	Kolkata, India	Manufacturer	Evolving Players	
Exfire	Jaypee Safety Systems Private Limited	Coimbatore, India	Manufacturer	Evolving Players	
Spectra Fire Systems	Spectra Fire Systems Pvt. Ltd.	New Delhi, India	Manufacturer	Evolving Players	
Vighnaharta	Vighnaharta Technologies Private Limited	Pune, India	Manufacturer	Evolving Players	
Kinde Fire	Add Mechatronics	Ahmedabad, India	Manufacturer	Evolving Players	
Firecore	Firecore Safety Engineers Pvt. Ltd.	Noida, India	Manufacturer	Evolving Players	
NOTOFIRE	NOTOFIRE PVT LTD	Noida, India	Manufacturer	Evolving Players	
UFS	Unicare Fire Safety (India) Pvt. Ltd.	Noida, India	Manufacturer	Evolving Players	
Enersafe	Enersafe Industry	Ahmedabad, India	Manufacturer	Evolving Players	
Avinash Fire	Avinash Fire Protection Systems Pvt. Ltd.	Ludhiana, India	Manufacturer	Evolving Players	
Ignite	Ignite Safety Solutions	Gurugram, India	Manufacturer	Evolving Players	
Hi-Care Safety Solutions	Hicare Protective Wear Pvt. Ltd.	Navi Mumbai, India	Manufacturer	Evolving Players	
SEMPROS	Sempros Fire N Safety Systems	Pune, India	Manufacturer	Evolving Players	
Vardan	Vardan Safety Solutions	Ahmedabad, India	Manufacturer	Evolving Players	

FireRID	Bluescope AV Technologies Pvt. Ltd.	Mumbai, India	Manufacturer	Evolving Players	
Marichi Fire	Marichi Fire And Safety	Ahmedabad, India	Manufacturer	Evolving Players	
IGNIX	Incenditech Pvt Ltd	Pune, India	Manufacturer	Evolving Players	
BRIO	Brio Fire Safety Device	New Delhi, India	Manufacturer	Evolving Players	
Aankur	Aankur Fire Systems And Consultants	Faridabad, India	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Sankar Advanced Mobility Solutions Pvt. Ltd. Sankar Mobility ● Leaders X 99 Y 93 Overall 96

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 100/100

> Sankar Advanced Mobility Solutions holds a clear, third-party verified capability (UL 2596) in flame-retardant and thermal runaway shielding materials used within advanced battery packs, though its focus remains on protective materials rather than pioneering novel, inherently safe core battery cell chemistries.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) manufactures "SanMica" barriers, silicone ceramic foams, and flame-retardant materials designed for battery pack protection, achieving a verified UL 2596 certification from a U.S. laboratory for its thermal runaway containment materials.
- SAMS has established an in-house thermal runaway testing facility and co-develops phase change materials (PCM) with Pluss Advanced Technologies, focusing on cell-to-pack safety insulation rather than developing proprietary internal cell chemistries.

Thermal Runaway Prevention & Suppression Systems — scored 96/100

> The company shows a clear, verifiable capability in passive thermal runaway containment and mitigation through its specialized mica products and in-house testing facilities, though it operates primarily as a component supplier rather than a leading-edge active system developer.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions Pvt. Ltd. offers SanMica, a line of mica-based barriers, flexible/rigid sheets, and 3D profiles engineered to act as passive protection against thermal runaway in electric vehicle battery packs.
- The company established an in-house thermal runaway testing facility in August 2024, and its SanMica barriers have been commercially ordered and qualified by a major U.S. company and a global industry leader.

Battery Management System (BMS) Safety Features — scored 100/100

> The company primarily markets and has gained certifications, such as UL 2596, for passive thermal runaway insulation and heat-shielding materials, rather than showcasing proprietary algorithms, sensor accuracy, or verified real-time electronic fail-safe response data for an advanced active Battery Management System.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) lists a "Battery Management System" and EV components like "charging units BMS etc." under its broader structural and thermal product portal, but its primary, proprietary safety innovations are explicitly focused on mechanical and passive thermal protection layers like SANMICA barriers, 3D Mica, and phase-change materials.

Compliance with Indian & International Safety Standards — scored 100/100

> SAMS shows clear, verifiable international capabilities through a specialized UL certification and an in-house thermal testing facility, a verified capability that still falls short of leading-edge third-party system certifications.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions Pvt. Ltd. (SAMS) has received official UL 2596 certification from a US-based laboratory for its thermal runaway protection materials.
- The company has established an indigenous, in-house thermal runaway testing facility to evaluate and validate its advanced safety materials, including its proprietary SANMICA thermal management barrier.

R&D and Innovation in Battery Fire Safety — scored 100/100

> The company demonstrates strong, third-party verified capabilities and dedicated testing infrastructure for battery safety components, a clear and verifiable capability.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) developed proprietary battery safety solutions including SanMica mica barriers and carbon fiber laminates, established an in-house thermal runaway testing facility, and achieved UL 2596 certification for its thermal-runaway-protection materials.
- SAMS partnered with Pluss Advanced Technologies Ltd. to co-develop innovative phase-change materials (PCM) that actively dissipate heat to mitigate battery thermal runaway risks.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 100/100

> Secondary research confirms that the company has extensive active commercial supply relationships covering the majority of India's electric two-wheeler market, but specific client names, contract valuations, and explicit agreement durations remain confidential and could not be independently audited through public third-party documents. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) secured its first commercial order for MICA thermal runaway protection in January 2023 and followed it with a supply order from a leading Indian electric scooter manufacturer in July 2023, according to the official SAMS Corporate Timeline.
- Executive disclosure by the company's CEO confirms that SAMS serves approximately 70% of India's electric two-wheeler OEMs with its battery thermal runaway protection and insulation systems, alongside gaining multiple global OEM approvals, as stated on the SAMS CEO Professional Profile.

Revenue & Growth in Indian Market — scored 95/100

> Overall revenue limits were verified, but specific segmented sales splits for fire safety products in the Indian market remained unverified due to the company's private status, leaving the evidence partial. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions Private Limited generated less than ₹10 Crore in operating revenue with a 12% CAGR based on corporate intelligence records.
- The company commercialized SANMICA thermal runaway protection and secured initial orders from EV scooter manufacturers according to corporate milestones.

Manufacturing Capacity & Localization in India — scored 96/100

> While the local manufacturing footprint and OEM supply reach are verified through parent group infrastructure, specific capacity figures in GWh and formal PLI scheme commitments could not be confirmed through secondary research due to the company's private entity status, leaving it a moderate performer. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) has localized production capabilities supplying thermal runaway protection and insulation systems to about 70% of India's electric two-wheeler OEMs.
- SAMS operates as a division of the Sankar Group, leveraging five domestic manufacturing plants across India, including facilities in Chennai, Pune, Tiruvanandapuram, and Uttarakhand.

After-Sales Service & Support Network — scored 81/100

> Secondary research confirms that the company has a physical footprint in Chennai and Bengaluru as well as an in-house technical testing facility for safety validation. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sankar Advanced Mobility Solutions (SAMS) lists dedicated corporate locations consisting of its primary manufacturing and corporate footprint in Chennai, Tamil Nadu, along with a regional office in Bengaluru opened in November 2023.
- As a tier-1 component vendor supplying passive materials like SanMica to approximately 70% of India's electric two-wheeler OEMs, the company's aftermarket footprint is institutional, anchored by an in-house thermal runaway testing facility established in August 2024 to support OEM engineering requirements and technical validation.

Regulatory Approvals & Certifications — scored 92/100

> International testing compliance (UL 2596) and commercial qualifications were verified via corporate disclosures, but based on primary research, the company has not disclosed specific certification numbers, approval dates, or formal compliance records from Indian regulatory authorities like ARAI, ICAT, or BIS, leaving the evidence partial and self-reported.

SUPPORTING POINTS

- The company's flagship thermal runaway protection material, SANMICA, successfully cleared international UL 2596 standard testing in the United States, as reported by company leadership.
- SANMICA was formally qualified by a major U.S. company in March 2024 and further qualified by a global industry leader in February 2025 following a product approval by a major electronics brand in December 2024.

Bharati Fire Engineers Bharati Fire ● Leaders X 87 Y 92 Overall 90

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 68/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Bharati Fire Engineers operates as a downstream fire safety equipment manufacturer specializing in systems like FireFite Lithex, rather than developing internal cell chemistries or pack-level flame-retardant materials.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 100/100

> Bharati Fire Engineers shows clear, third-party verified, and commercially adopted active suppression capabilities specifically engineered for lithium-ion thermal runaway mitigation through its licensed AVD technology integration.

SUPPORTING POINTS

- Bharati Fire Engineers manufactures FireFite Lithex, a specialized fire suppression technology designed in collaboration with AVD (UK) utilizing Aqueous Vermiculite Dispersion (AVD). It provides both fixed and mobile delivery methods to successfully control, cool, and extinguish lithium-ion battery fires and thermal runaway events.
- The company's battery fire suppression products are BIS-approved (ISI mark) and have been tested or integrated by national R&D organizations like DRDO and CFEES, as well as primary EV manufacturers such as Tata Motors and Ola Electric.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Bharati Fire Engineers focuses strictly on downstream active fire suppression, detection systems, and specialized extinguishing agents—such as its AVD-based FIREFITE-LITHEX units—and offers no native Battery Management System (BMS) software, real-time safety algorithms, or preventative electronics.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Compliance with Indian & International Safety Standards — scored 98/100

> The company holds clear, verifiable compliance and certifications (BIS/ISI mark, AIS-0135, and UL-certified components) tailored for EV fire safety and suppression, though it does not explicitly demonstrate leading-edge system-level certifications like full-scale IEC 62619 or UL 1973 battery safety alignment.

SUPPORTING POINTS

- Bharati Fire Engineers holds BIS approval (ISI mark) for its "FireFite Lithex" extinguisher, which is designed specifically for lithium-ion battery fires using aqueous vermiculite dispersion (AVD) technology.
- The company's automobile detection and suppression system (FDSS) for vehicles, including electric buses, is certified according to AIS-0135 and features components that are UL listed.

R&D and Innovation in Battery Fire Safety — scored 95/100

> The company shows clear, verifiable deployment capabilities through international partnerships, although foundational R&D was conducted externally.

SUPPORTING POINTS

- Bharati Fire Engineers launched the FIREFITE LITHEX extinguisher utilizing Aqueous Vermiculite Dispersion (AVD) technology to combat lithium-ion battery fires.
- The company secured and licensed this patented technology from a UK-based corporation for production under the Make in India initiative.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 100/100

> Verified named customer deployments and regulatory compliance exist, but specific contract values and durations are undisclosed, leaving the evidence partially reported.

SUPPORTING POINTS

- Press releases and corporate documentation note that Bharati Fire Engineers' FireFite Lithex line has been deployed or referenced regarding major Indian EV manufacturers such as Tata and Ola.
- Company documentation details AIS-013S certified and SP-approved automobile fire detection and suppression systems applied in heavy transport and electric utility segments.

Revenue & Growth in Indian Market — scored 95/100

> Overall corporate revenue and total growth were verified through official registry platforms, but based on primary research, the company has not disclosed isolated sales figures for battery safety products, leaving the evaluation partial.

SUPPORTING POINTS

- Official registry records via corporate research platforms show that Bharati Fire Engineers Private Limited generated an operating revenue in the range of ₹1 Crore to ₹10 Crore for the financial year ending March 31, 2025, reflecting significant growth following its July 2023 incorporation.
- Public filings and industry portfolios indicate the company offers specialized lithium-ion battery fire safety solutions like "FireFite-Lithex" alongside general fire safety equipment, though public financial documentation lacks a segmented breakdown isolating the specific revenue derived exclusively from battery fire safety features.

Manufacturing Capacity & Localization in India — scored 88/100

> Secondary research confirms that the company has a physical manufacturing facility and full localization within India, in Navi Mumbai, but based on primary research, the company has not disclosed any GWh battery fire safety production capacity figures or localization commitments under the PLI scheme, since the firm functions as a component fabricator rather than a cell or pack assembler.

SUPPORTING POINTS

- Bharati Fire Engineers operates a local manufacturing facility (Plant) spanning approximately 25,000 square feet located at Plot No. A-427 & A-428, TTC Industrial Estate, MIDC Mahape, Navi Mumbai, India, which houses their production and testing machinery.
- According to the Ministry of Corporate Affairs (MCA) records, the entity was formally incorporated as Bharati Fire Engineers Private Limited on 24 July 2023 with its registered office in Mumbai, Maharashtra, to engage in the manufacture of machinery, equipment, and structural metal products.

After-Sales Service & Support Network — scored 79/100

> The company's self-reported warranty provision was verified, but the exact number of distributed service points across India, quantifiable response times, and third-party verified customer satisfaction metrics could not be confirmed through secondary research. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Bharati Fire Engineers self-reports standard after-sales maintenance and warranty support directly through its centralized technical staff and manufacturing facility operations, as outlined in the official FireFite Quality Assurance Statement.
- Corporate registration and public location records indicate the firm's physical infrastructure is localized within the Mumbai Metropolitan Region, specifically consisting of its corporate headquarters in Mulund West and its manufacturing facility in Mahape, Navi Mumbai, according to Justdial Location Records.

Regulatory Approvals & Certifications — scored 100/100

> Secondary research confirms the company's compliance with Indian automotive regulations (AIS:0135, AIS:004) and BIS/ISI markings for its specialized battery safety line, as well as international UL certification for its aerosol generators, though specific certificate numbers and official approval dates were not accessible through secondary research, leaving it in a clear capability position. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Bharati Fire Engineers states that its automotive FDSS (Fire Detection & Suppression System), which incorporates its aerosol generator and is applied to battery compartments, is approved and tested under Indian automotive standard AIS:0135, tested for EMI/EMC under AIS:004, and utilizes a UL Certified chemical and aerosol generator, according to the FireFite Automobile Product Page.
- The company's dedicated battery fire safety extinguisher line, FireFite Lithex (developed in collaboration with AVD, UK for lithium-ion battery fires), has officially received BIS approval and bears the ISI mark in India, as documented in an industry report by Ecargolog.

Pluss Advanced Technologies Pvt. Ltd. (subsidiary of Carborundum Universal Limited)

Pluss · Leaders

X 89 Y 87 Overall 88

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 90/100

> Pluss Advanced Technologies shows present but partial capability via self-reported thermal containment metrics without independent third-party certifications or safer raw chemistry development.

SUPPORTING POINTS

- Pluss Advanced Technologies utilizes proprietary "savE" Phase Change Materials (PCMs) for passive thermal management around lithium-ion cells, which are self-reported to delay thermal runaway propagation by 3 to 4 times and maintain peak temperatures at 46°C.
- The company focuses on passive thermal buffer materials rather than alternative cell chemistries like solid-state or LFP, or structural flame-retardant pack components.

Thermal Runaway Prevention & Suppression Systems — scored 93/100

> Pluss Advanced Technologies shows a clear, verifiable capability in passive thermal runaway prevention through its specialized phase change materials, although its public documentation provides self-reported performance figures rather than extensive independent third-party certifications.

SUPPORTING POINTS

- Pluss Advanced Technologies provides passive Battery Thermal Management Systems (BTMS) leveraging Phase Change Materials (PCMs) that absorb substantial heat fluxes to regulate cell uniformity and delay thermal runaway time by three-fold.
- The company reports that its PCM-based battery jackets limit cell-to-cell heat propagation in the event of a fire, reducing the instance of a thermal runaway event by 4 times and delaying the chain reaction by 40 minutes compared to 5 to 10 minutes without PCMs.

Battery Management System (BMS) Safety Features — scored 78/100

> The company's core technological focus is entirely on material-based passive thermal energy storage (PCMs) rather than proprietary active BMS software, safety monitoring algorithms, or electronic fail-safe mechanisms.

SUPPORTING POINTS

- Pluss Advanced Technologies Pvt. Ltd. is a material science company that primarily manufactures phase change materials (PCMs) for passive thermal management and thermal runaway mitigation, rather than active electronics.
- Public commentary and industry literature note that Pluss's passive PCM materials are designed to be coupled alongside external or third-party Battery Management Systems (BMS) to enhance battery pack life and safety.

Compliance with Indian & International Safety Standards — scored 89/100

> The company shows awareness and compliance intent regarding standards like AIS-156, but lacks publicly available third-party certifications or test reports for international benchmarks such as IEC 62619 or UL 1973. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Pluss Advanced Technologies' leadership publicly addresses India's AIS-156 battery safety regulations regarding the prevention of EV fires and management of thermal runaway using phase-change materials.
- The company provides proprietary savE® PCM specifications and datasheets while noting the critical importance of adhering to relevant sector regulations.

R&D and Innovation in Battery Fire Safety — scored 96/100

> The company shows a clear, verifiable passive thermal safety capability backed by an officially recognized DSIR laboratory and concrete performance figures, though its primary industry footprint lies in broad temperature control rather than exclusively leading-edge active fire suppression systems.

SUPPORTING POINTS

- Pluss Advanced Technologies Pvt. Ltd. has developed specific passive battery thermal management systems (BTMS) using its proprietary "savE®" phase change materials (PCMs) and form-stable PCM blocks, which are engineered to reduce the occurrence of thermal runaway by four times and delay cell-to-cell propagation reactions by 40 minutes.
- The company maintains a dedicated in-house research arm (CREST) and a state-of-the-art research laboratory officially recognized by the Government of India's Department of Scientific and Industrial Research (DSIR), supporting their ongoing proprietary patents and R&D pipelines for specialized thermal energy storage materials.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 86/100

> Secondary research confirms that Pluss manufactures materials aimed at battery safety, but based on primary research, the company has not disclosed any named Tier-1 OEM contracts, financial contract values, or duration milestones specific to the advanced battery fire safety industry, given a lack of public disclosure.

SUPPORTING POINTS

- Pluss Advanced Technologies highlights its savE® line of phase change materials for passive battery thermal management and thermal runaway mitigation in electric vehicles. Based on primary research, the company has not disclosed further details.
- Outside of battery fire safety, the company actively demonstrates third-party partnerships in other applications, such as a major joint initiative with Ecodome Logistics to distribute reusable, IoT-enabled PCM cold-chain boxes across India.

Revenue & Growth in Indian Market — scored 94/100

> Total corporate revenue boundaries and product capabilities are verified, but specific Indian market sales revenue and growth rates for battery fire safety products are absent, leaving the evaluation present but partial.

SUPPORTING POINTS

- Pluss Advanced Technologies Limited's overall company operating revenue range is reported between ₹100 Cr and ₹500 Cr for the financial year ending March 31, 2026, as verified by third-party corporate data filings.
- The company officially manufactures a proprietary line of chemical and bio-based savE® phase change materials explicitly engineered for passive lithium-ion battery thermal management and mitigating thermal runaway risks.

Manufacturing Capacity & Localization in India — scored 85/100

> Secondary research confirms the company's local manufacturing presence in Bawal and R&D operations in Gurugram. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Pluss Advanced Technologies operates a manufacturing facility in Bawal, Haryana, India, for fabricating its specialty polymeric additives and Phase Change Materials (PCMs).
- The company maintains its corporate headquarters in Gurugram, Haryana, along with an R&D and engineering scale-up division via the Centre for Research in Energy Science & Technology (CREST).

After-Sales Service & Support Network — scored 77/100

> The company's maintenance-free engineering model and central headquarters were verified, but based on primary research, the company has not disclosed data on dedicated service centers, response times, or customer satisfaction metrics across India.

SUPPORTING POINTS

- Pluss Advanced Technologies states that its savE phase change materials operate via a self-contained chemical phase change with zero moving components, pumps, or seals, rendering the core passive thermal management technology maintenance-free over its operational lifespan.
- Technical support and corporate operations are coordinated from its headquarters in Gurugram, India, alongside international hubs in the Netherlands, United States, Thailand, and Turkey.

Regulatory Approvals & Certifications — scored 92/100

> Secondary research confirms the company's international material quality certifications and facility ISO compliance, but based on primary research, the company has not disclosed specific battery fire safety certificate numbers, approval dates, or formal compliance clearances from Indian automotive regulatory bodies like ARAI, ICAT, or BIS for its battery thermal management line.

SUPPORTING POINTS

- Pluss Advanced Technologies operates DSIR-certified R&D centers and manufacturing facilities that are audited and certified under ISO 9001:2015 and ISO 13485 quality management standards.
- The company's savE OM50 phase change material is explicitly certified by the RAL Quality Association PCM under the strict third-party RAL Quality Mark PCM standard.

S-Square Enterprises ● Leaders

X 85 Y 89 Overall 87

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 100/100

> S-Square Enterprises shows a clear capability in producing fire-resistant and flame-retardant materials for cell and pack design to reduce fire risk, though its public profiles do not showcase third-party certified chemistry performance, such as nail penetration or overcharge safety data, on completed cell assemblies.

SUPPORTING POINTS

- S-Square Enterprises manufactures UL94V-0 grade flame-retardant materials, such as polycarbonate thermal cell spacers, MPP foam, CR, EPDM, and Nitrile insulation foams for battery packs.
- S-Square Enterprises manufactures flame-retardant PP, PVC, and FRPC insulation components, battery thermal pads, and aerogel insulation engineered for electric vehicles and Battery Energy Storage Systems (BESS).

Thermal Runaway Prevention & Suppression Systems — scored 95/100

> S-Square Enterprises shows a clear, verifiable passive capability through manufacturing specialized battery thermal barrier pads, aerogel, and flame-retardant insulation components designed specifically to mitigate cell-to-cell thermal runaway propagation in EV and BESS environments, though third-party testing documentation or active suppression product integrations are not prominently detailed.

SUPPORTING POINTS

- S-Square Enterprises manufactures EV battery insulation materials, including thermal barriers between individual cells and thermal insulation within the battery pack designed to slow heat transfer and confine thermal runaway events.
- The company delivers high-performance EV battery insulation and flame retardant (FR) foams, supplying products such as engineered FR PP, PVC, FRPC insulation, battery thermal barrier pads, aerogel, and precision custom foam die-cuts for EV and BESS systems.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. A comprehensive search reveals that S-Square Enterprises focuses strictly on manufacturing passive physical insulation materials rather than electronic Battery Management Systems (BMS) with real-time monitoring.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Compliance with Indian & International Safety Standards — scored 70/100

> The company operates primarily as a material supplier rather than a system integrator with verifiable battery safety standard compliance. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- S-Square Enterprises is an IATF 16949 certified manufacturer specializing in EV battery insulation solutions and compliance decals.
- Based on primary research, the company has not disclosed information for this parameter.

R&D and Innovation in Battery Fire Safety — scored 89/100

> The company shows present but partial, self-reported manufacturing and customization capabilities for battery thermal barriers, but lacks any third-party verified R&D metrics, academic research collaborations, or patent evidence indicating leading-edge innovation. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- S-Square Enterprises self-reports as an IATF 16949 certified manufacturer that designs custom materials, such as cell-to-cell thermal barriers, thermal runaway protection pads using aerogel and mica, and flame-retardant foam gaskets specifically engineered to control heat propagation in electric vehicles and battery energy storage systems (BESS).
- The company publicizes that it collaborates closely with battery pack makers and automobile manufacturers to achieve specific functional requirements for its thermal insulation layouts, though it discloses no formal patent filings, explicit R&D spend metrics, or published scientific papers.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 84/100

> The company self-reports general engineering collaborations with automotive and battery manufacturers, but based on primary research, the company has not disclosed specific evidence of major Tier-1 OEM contracts, contract values, or named agreements.

SUPPORTING POINTS

- S-Square Enterprises states on its official corporate platform that it manufactures customized EV battery insulation, thermal pads, and aerogel fire barriers, collaborating directly with unnamed battery pack makers and automobile manufacturers to meet precise design and technical specifications.
- According to primary research, public commercial registries and business profiles report that S-Square Enterprises operates as a small enterprise with up to 10 employees, without any third-party public disclosure or documentation regarding named OEM clients, agreement durations, or contract values.

Revenue & Growth in Indian Market — scored 94/100

> The company's regional supply presence and specialized fire safety material lines, such as FR-PP, FR-PC, and aerogel barriers, were verified through its official corporate catalogue. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- S-Square Enterprises publicly documents its regional manufacturing and local supply capabilities tailored for electric vehicle (EV) battery packs and Battery Energy Storage Systems (BESS) enclosures within key industrial hubs around Pune, Chakan, Pimpri-Chinchwad, Bhosari, Thane, and Mumbai.
- S-Square Enterprises self-reports over 20 years of manufacturing experience across industrial labels, decals, gaskets, and vacuum forming parts, alongside its expanding portfolio of UL94 V-0 flame-retardant certified materials, thermal pads, and aerogel insulation.

Manufacturing Capacity & Localization in India — scored 92/100

> S-Square Enterprises has verified local manufacturing operations and custom component fabrication within India for EV insulation materials. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- S-Square Enterprises hosts local design, supply, and precision die-cutting fabrication across industrial hubs in Pune, India, supplying specialized EV battery insulation materials, aero-gels, and flame-retardant barriers directly to regional battery pack makers and automotive OEMs.
- The company maintains localization capabilities extending directly across major Maharashtra manufacturing clusters, with dedicated commercial localized supply frameworks built out for OEMs in Pune, Pimpri-Chinchwad, Bhosari, and Thane.

After-Sales Service & Support Network — scored 87/100

> Localized pre-sales technical support and regional facility proximity are self-reported, but based on primary research, the company has not disclosed a formal pan-India network of service centers, quantified response times, and structured warranty management systems, leaving the evidence strictly partial and self-reported.

SUPPORTING POINTS

- S-Square Enterprises self-reports providing "Responsive Engineering Support," detailing close collaboration with client technical teams through material design, prototyping, and compliance validation processes as documented on the S-Square Enterprises EV Battery Insulation Service Page.
- The company showcases regional hub capabilities via its facility in the Mathura Industrial Area, offering factory pick-ups and dedicated on-site technical visits for neighboring industrial clusters as detailed in its Bhosari MIDC Hub Capability Overview.

Regulatory Approvals & Certifications — scored 90/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- S-Square Enterprises publicly highlights its status as an IATF 16949 Certified Manufacturer for high-performance EV battery insulation and flame-retardant foams.
- The company self-reports compliance metrics for its custom insulation materials (FR-PP, FR-PC, and FR-PVC), certifying them to meet international UL94 V-0 / VTM-0 flammability standards alongside RoHS and REACH standards.

Lion Battery ● Leaders

X 85 Y 87 Overall 86

X — PRODUCT CAPABILITY

Thermal Runaway Prevention & Suppression Systems — scored 87/100

> Lion Battery shows present but basic passive mitigation capabilities through its self-reported line of insulated metal storage and charging boxes. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lion Battery based in Umargam, Valsad, Gujarat self-reports that they manufacture "LiPo Safe Metal Boxes" and "Lithium Safe Battery Boxes" featuring heavy-duty steel construction with a heat-resistant insulated interior lining intended to reduce risks during battery storage, charging, and transportation.
- Lion Battery's public documentation details physical enclosure integration, custom carbon fiber boxes, and general thermal considerations, but lacks references to integrated active suppression agents, automated venting valves, or third-party verified fire-propagation test results for these safety containment boxes.

Fire-Resistant Battery Chemistry & Materials — scored 68/100

> Secondary research confirms that the company manufactures external fire-resistant structural boxes and containers. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lion Battery focuses on manufacturing structural storage containers and custom enclosures from heavy-duty powder-coated steel rather than chemistry modifications, noting that their carbon-fiber and metal safety boxes are designed externally to help contain battery-related incidents rather than acting as a cell insulation material.
- Product specifications from Lion Battery for their custom battery box units show that while optional internal fire-resistant insulation layers can be added to the metal box structures, they do not produce inherently safe internal battery chemistries, flame-retardant cell separators, or solid-state cell architecture.

Battery Management System (BMS) Safety Features — scored 96/100

> Basic integration and repair capabilities were verified, but specific technical details on proprietary advanced monitoring algorithms, sensor accuracies, or safety-critical response times were completely unavailable.

SUPPORTING POINTS

- Lion Battery integrates baseline "Smart BMS" configurations into its custom lithium-ion, NMC, and LiFePO4 battery packs built at its manufacturing infrastructure in GIDC Umargam, Valsad, as stated on the official Lion Battery Home Page.
- The company offers specialized services including "BMS integration" and "BMS repair" alongside cell matching and final quality testing at its Unit-1 and Unit-2 facilities in Gujarat, according to its Lion Battery Manufacturing Process Guide.

Compliance with Indian & International Safety Standards — scored 87/100

> Secondary research confirms that the company sells components targeting India's AIS-156 standards and self-reports fire-resistant manufacturing capabilities for its box containers. Based on primary research, the company has not disclosed any independent, third-party laboratory test certification reports matching global standards like UL 1973 or IEC 62619 for the physical safety boxes themselves.

SUPPORTING POINTS

- Lion Battery's Official E-Commerce Store lists a "Smart-BMS-LFP-20S-80A-AIS156-Daly" component, proving the sale of specific battery management systems engineered for AIS-156 automotive compliance within India.
- Lion Battery's Product Documentation for the Lithium Safe Battery Box (300×200×200mm) states that their fireproof storage containers are insulated containers designed internally to mitigate thermal runway events, but no explicit third-party test certifications under IEC 62619, UL 1973, or UN38.3 are publicly published for their safety box enclosures.

R&D and Innovation in Battery Fire Safety — scored 89/100

> Secondary research confirms that the company designs and produces insulated flame-retardant enclosures and evaluates structural material safety parameters for custom applications. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lion Battery utilizes a specific heat-resistant insulated interior lining and a heavy-duty powder-coated steel outer construction for their custom-sized LiPo Safe Box models to mitigate thermal events during storage and charging, as documented on the Lion Battery Product Catalog.
- Lion Battery expanded their engineering design capabilities by introducing a custom carbon fiber battery box and enclosure service in Umargam, Gujarat, which integrates physical thermal management, mechanical protection, and fire safety design parameters specific to EV, drone, and robotics applications, as outlined on the Lion Battery Carbon Fiber Service Page.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 74/100

> The company primarily self-reports a readiness to provide custom OEM/ODM packaging and generic metal battery safety boxes, but lacks public verification of active supply contracts with established Tier-1 automotive or electronics OEMs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lion Battery's promotional materials indicate it offers custom lithium battery pack design, manufacturing, and OEM/ODM partnerships from its Gujarat facility, catering to startups, system integrators, and industrial OEMs.
- Based on primary research, the company has not disclosed evidence regarding signed supply agreements, contract values, durations, or named tier-1 Indian electric vehicle manufacturers or energy storage integrators utilizing their fire safety box or battery enclosure solutions.

Revenue & Growth in Indian Market — scored 87/100

> The product "Lion-Ex" exists as a verifiable technical offering in the Indian battery fire safety market. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lifeguard India offers "Lion-Ex," marketed as India's first indigenous lithium-ion battery fire extinguisher utilizing a specialized Hydrated Mica Suspension (HMS) agent designed to mitigate thermal runaway.
- Based on primary research, the company has not disclosed evidence of standalone sales revenue or isolated growth rates for the "Lion-Ex" product line could be located in financial statements or investor reports.

Manufacturing Capacity & Localization in India — scored 92/100

> The company demonstrates verified local pack assembly capabilities for custom applications in India. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lion Battery operates local pack assembly in India, assembling custom Li-ion, NMC, and LiFePO4 battery packs ranging from 2Ah up to 15kWh using imported cells from trusted brands like Samsung, LG, and Panasonic.
- Based on primary research, the company has not disclosed evidence indicating any large-scale gigawatt-hour (GWh) cell manufacturing capacity or official localization commitments under India's Production Linked Incentive (PLI) schemes.

After-Sales Service & Support Network — scored 91/100

> Lion Battery shows localized capabilities through a dedicated repair facility and general claims of aftermarket support, yet it lacks external third-party verification or evidence of a widespread nationwide support network in the fire-safety storage space, leaving the evidence present but partial and self-reported.

SUPPORTING POINTS

- Lion Battery lists basic after-sales service metrics on its site, claiming fast turnaround for repairs and custom builds alongside friendly customer support and an on-site professional repair service centre for Li-ion, LiFePO4, and NMC packs.
- Publicly verifiable nationwide metrics, exact network sizing, response time performance logs, or third-party validated customer satisfaction data across India are absent from the open market, with their core physical facility based in Umargam, Gujarat.

Regulatory Approvals & Certifications — scored 92/100

> The company holds verifiable historical third-party BIS certifications and claims adherence to current safety standards, but lacks detailed, up-to-date public evidence of active ARAI/ICAT approvals or explicit current certificate numbers across its product range.

SUPPORTING POINTS

- Lion Battery Systems (Shanghai) Ltd. holds a Bureau of Indian Standards (BIS) CRS registration under certificate number R-41021695.
- Lion Battery India self-reports compliance with mandatory Indian BIS certifications, UN38.3 transportation safety, and IEC 62133 safety standards, though specific certificate numbers for its local pack assemblies are not openly listed.

AGNI Devices Pvt. Ltd. AGNi Challengers X 72 Y 88 Overall 80

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 67/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. AGNI Devices Pvt. Ltd. operates as a manufacturer of conventional fire alarm systems and external suppression equipment rather than an advanced battery developer, and based on primary research, the company has not disclosed fire-resistant battery chemistries or cell-level pack materials.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 79/100

> The company has clear capabilities in traditional enclosure fire suppression, but it provides no specialized or third-party verified solutions dedicated specifically to advanced battery systems and thermal runaway events. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- AGNI Devices Pvt. Ltd. manufactures general fire protection equipment, including the AGNi Advanced Fire Suppression System featuring fast-response tubing and automatic extinguishing panels for critical enclosures.
- Based on primary research, the company has not disclosed evidence or third-party test outcomes were located linking AGNI Devices' products specifically to lithium-ion battery thermal runaway prevention, mitigation, or specialized advanced battery cooling/venting systems.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. AGNI Devices Pvt. Ltd.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Compliance with Indian & International Safety Standards — scored 69/100

> The company demonstrates verified compliance with conventional commercial life safety standards, but the complete absence of verifiable metrics or certifications for advanced battery and electric vehicle fire protection limits its standing here. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- AGNI Devices Pvt. Ltd. holds standard certifications including LPCB, CE, BIS, CPRI, and ERTL approvals for its commercial fire alarms and extinguishers.
- Based on primary research, the company has not disclosed evidence for compliance with dedicated battery fire safety standards such as AIS-156, IEC 62619, UL 1973, or UN38.3.

R&D and Innovation in Battery Fire Safety — scored 72/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. The complete absence of verifiable R&D metrics, patents, or institutional collaborations limits the company's standing on innovation capability.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence regarding dedicated R&D expenditure, proprietary battery fire safety patents, or advanced material testing collaborations for AGNI Devices Pvt. Ltd.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 76/100

> Secondary research looked extensively for EV OEM or energy storage contracts. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed further details.
- A comprehensive search of public enterprise records and tender listings demonstrates that the company's verified client base remains anchored in traditional government infrastructure and building safety projects rather than advanced lithium-ion battery OEM supply chains.

Revenue & Growth in Indian Market — scored 97/100

> Secondary research confirms the overall company revenue and broad growth rate through third-party financial aggregators. Based on primary research, the company has not disclosed specific sales metrics or growth rates exclusively tied to the company's lithium-ion battery fire safety product segment versus its conventional business lines.

SUPPORTING POINTS

- According to a company intelligence report from Tracxn, AGNI Devices Private Limited generated an annual operating revenue of ₹66.2 Crore for the financial year ending on March 31, 2025.
- Corporate filing aggregates from EMIS and Tracxn reveal that AGNI Devices Pvt. Ltd. experienced a compounded annual growth rate (CAGR) of 13% in revenue over the last year, with total operating revenue climbing into the ₹50 Cr – ₹100 Cr range.

Manufacturing Capacity & Localization in India — scored 86/100

> Secondary research confirmed the company's local manufacturing presence via its Sonepat facility and its expansion into Class L battery fire extinguishers. Based on primary research, the company has not disclosed such capacity data.

SUPPORTING POINTS

- AGNI Devices Pvt. Ltd. operates a primary manufacturing facility located at Plot No. 203, Phase IV, Sector 57, HSIIDC Industrial Estate, Kundli, Sonepat, Haryana, alongside offices in the Keshopur Industrial Area, New Delhi.
- The company manufactures traditional and advanced fire protection ecosystems, including specialized Class L (Lithium fire) extinguishers and tube suppression systems designed specifically for lithium battery risk environments.

After-Sales Service & Support Network — scored 91/100

> Core warranty terms and basic technical desk channels are verified, but specific metrics on response times, quantified numbers of dedicated service centers, or customer satisfaction logs were not accessible through secondary research, leaving the evidence at a baseline, self-reported level. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- AGNI Devices Pvt. Ltd. provides a standard manufacturer's warranty on its fire safety equipment portfolio, supported by a central technical, logistical, and sales desk operating via their New Delhi headquarters.
- Outside its direct headquarters, the company relies on a local dealer and distributor network across different regions like Pune to manage distributed equipment supply, installation assistance, and maintenance services.

Regulatory Approvals & Certifications — scored 90/100

> Secondary research confirms overall institutional compliance (BIS, LPCB, CE, ERTL) for Agni's foundational fire alarms and legacy fire extinguishers, but based on primary research, the company has not disclosed third-party certifications, certificate numbers, or approval dates specifically from automotive or advanced test bodies like ARAI or ICAT for its lithium-ion tube suppression systems or specialized Class L extinguishers.

SUPPORTING POINTS

- Agni Devices Pvt. Ltd. holds Bureau of Indian Standards (BIS) and ISI-marked certifications (IS 15683) for its general range of conventional fire extinguishers, such as its BC Type Stored Pressure Fire Extinguisher.
- The company maintains standard organizational and general fire alarm product line approvals from third parties, including LPCB, CE, ERTL (North) tested compliance for conventional panels, and an ISO 9001:2015 accreditation.

NewAge Fire Protection Industries Pvt. Ltd. NewAge Challengers X 72 Y 87 Overall 80

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. NewAge Fire Protection Industries Pvt. Ltd.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 70/100

> NewAge Fire Protection Industries Pvt. Ltd. is an established traditional fire safety manufacturer. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence for dedicated thermal runaway prevention or lithium-ion battery fire suppression systems manufactured by the company.
- The company's standard industrial offerings focus heavily on traditional firefighting equipment, including hydrants, water mist systems for turbine rooms, and clean agent systems for conventional cable vaults.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. The company specializes in traditional firefighting hoses, hydrants, valves, and building-level fire alarm control panels that offer BMS integration protocol outputs, rather than advanced battery management systems for managing thermal runaway within lithium-ion cells.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence indicating that NewAge Fire Protection Industries Pvt. Ltd. develops or provides advanced Battery Management System (BMS) safety features, algorithms, or real-time internal cell monitoring.

Compliance with Indian & International Safety Standards — scored 81/100

> The company manufactures lithium-ion battery fire extinguishers and hardware, but its public evidence lacks specific verification or type-approval under explicit advanced battery testing standards like AIS-156 or UL 1973. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- NewAge Fire Protection Industries Pvt. Ltd. showcases specialized hardware under its LIFEGUARD® brand tailored for EV, Lithium-Ion Battery Fire Safety, and BESS protection at major industry conventions.
- Based on primary research, the company has not disclosed evidence for direct third-party certifications or test reports specifically matching automotive/stationary battery safety standards like AIS-156, IEC 62619, UL 1973, or UN38.3, though general fire safety certifications are maintained.

R&D and Innovation in Battery Fire Safety — scored 72/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. NewAge Fire Protection Industries Pvt. Ltd.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 74/100

> The company manufactures fire protection systems applicable to technical environments, but based on primary research, the company has not disclosed active supply contracts, named clients, contract values, or agreement durations with major Indian electric vehicle manufacturers or energy storage integrators.

SUPPORTING POINTS

- NewAge Fire Protection Industries Pvt. Ltd. lists its primary system portfolio for complex industrial layouts in its official product brochures, including "Aerosol System Presentations" and a "NewAge NextGen" product line, targeted at advanced risk environments.
- The company's verified client sectors, published via its core business infrastructure channels, include heavy engineering, oil and gas, offshore marine operations, power plants, and major EPC contractors, rather than EV OEMs.

Revenue & Growth in Indian Market — scored 94/100

> The company's macro-level corporate revenue and overall growth rate were third-party verified, but based on primary research, the company has not disclosed specific revenue breakdowns and growth rates dedicated exclusively to battery fire safety products.

SUPPORTING POINTS

- Third-party registry reports from Tofler and Tracxn confirm that Newage Fire Protection Industries Private Limited operates in the operating revenue range of ₹100 cr - ₹500 cr for the financial year ending on March 31, 2025.
- Corporate research profiles from EMIS indicate that the entity achieved a net sales revenue increase of 10.14% in its overall 2025 financial highlights.

Manufacturing Capacity & Localization in India — scored 85/100

> Secondary research confirms that the company locally manufactures industrial fire safety equipment across multiple facilities in Maharashtra. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- ICRA credit rating reports verify that NewAge Fire Protection Industries Pvt. Ltd. maintains its primary manufacturing facility at Khopoli, Thane district in Maharashtra, India.
- Corporate profile filings document that the group operates five manufacturing premises in India producing engineering fire-safety equipment such as fire hoses, hydrants, valves, and water foam monitors.

After-Sales Service & Support Network — scored 91/100

> While the presence of local support channels, AMC capabilities, and an active internal service management team was verified, specific metrics regarding the exact quantity of regional service points, guaranteed breakdown response times, or third-party verified customer satisfaction results could not be confirmed through secondary research. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- NewAge Fire Protection Industries self-reports a business model of "Global Supply with Local Support," detailing a dedicated network of authorized distributors globally and across India for product installation, routine servicing, and field maintenance.
- Commercial service listings and recruitment data show that the company provides operational fire-fighting Annual Maintenance Contracts (AMC) and repair services managed out of their regional facilities, actively hiring for dedicated in-house service department roles such as a "Service Manager" based in Mumbai to orchestrate after-sales service operations.

Regulatory Approvals & Certifications — scored 90/100

> Secondary research confirms that NewAge holds extensive BIS compliance and global fire-safety listings (UL, FM) for conventional firefighting hardware. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- NewAge Fire Protection Industries Pvt. Ltd. holds formal Bureau of Indian Standards (BIS) certifications for its core equipment lineup, with specific products certified under Indian Standards such as IS 636 Type A, B, & C for its Pyroprotect fire hose lines and IS 5290 Type-A for its single hydrant valves.
- The company maintains international third-party quality and product safety validations, including an active ISO 9001:2015 Quality Management System status alongside hardware-level equipment listings from international bodies like Underwriters Laboratories (UL) and Factory Mutual (FM).

Hiren Industrial Corporation ● Challengers

X 78 Y 80 Overall 79

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 70/100

> The company manufactures downstream fire suppression systems rather than developing inherently safer battery chemistries or flame-retardant pack materials, resulting in a low assessment due to the absence of direct evidence for this specific measure. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation lists a "Lithium Battery Fire Extinguisher" featuring a fluorine-free water-additive/surfactant liquid designed for rapid heat absorption and thermal runaway cooling.

Thermal Runaway Prevention & Suppression Systems — scored 87/100

> The company offers a targeted, portable lithium-ion battery fire extinguisher product featuring a dedicated cooling chemistry, but it does not manufacture or integrate advanced, commercial-scale automated active or passive suppression or cooling systems.

SUPPORTING POINTS

- Hiren Industrial Corporation markets a specialized "Lithium Battery Fire Extinguisher" featuring a fluorine-free water-additive/surfactant liquid agent designed specifically for rapid heat absorption to achieve thermal runaway cooling and suppress gas release.

Battery Management System (BMS) Safety Features — scored 68/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation provides a "Lithium Battery Fire Extinguisher" containing a fluorine-free cooling and wetting surfactant liquid designed to suppress gas release and cool thermal runaway.
- Based on primary research, the company has not disclosed evidence regarding any Battery Management System (BMS) software, hardware, or electronic safety algorithms developed or sold by the company.

Compliance with Indian & International Safety Standards — scored 80/100

> The company offers specialized thermal runaway suppression equipment but lacks verifiable third-party test reports or compliance certificates for advanced battery safety standards. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation manufactures a lithium battery fire extinguisher utilizing a fluorine-free cooling water-additive agent designed for thermal runaway cooling and suppressing gas release.
- Based on primary research, the company has not disclosed evidence or third-party certifications regarding major battery safety standards such as AIS-156, IEC 62619, UL 1973, or UN38.3 were located for the company's product line.

R&D and Innovation in Battery Fire Safety — scored 83/100

> The company has an active commercial offering tailored to lithium-ion battery fire suppression, but its technical capabilities remain largely self-reported or unproven via external innovation metrics. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation markets a dedicated Lithium Battery Fire Extinguisher utilizing a fluorine-free cooling surfactant liquid designed to suppress gas release and manage thermal runaway.
- Based on primary research, the company has not disclosed evidence regarding proprietary patents, specific R&D expenditure figures, academic research collaborations, or third-party verified technological breakthroughs in advanced battery fire safety chemistry.

Y — BUSINESS CAPABILITY

Regulatory Approvals & Certifications — scored 82/100

> Secondary research confirms general BIS compliance for legacy products. Based on primary research, the company has not disclosed battery-specific regulatory approvals or certification numbers.

SUPPORTING POINTS

- Hiren Industrial Corporation's standard product listings reference compliance with Bureau of Indian Standards (BIS) parameters such as IS 15683 and general ISI marks for traditional lines like ABC Dry Chemical Powder.
- Documentation for the specialized lithium battery fire extinguisher product line outlines basic technical specifications and cooling mechanisms, but omits specific certificate numbers, approval dates, or compliance records from specialized bodies like ARAI, ICAT, or international organizations.

Tier-1 OEM Partnerships & Supply Contracts — scored 66/100

> Secondary research looked for verifiable tier-1 electric vehicle, energy storage, or electronics OEM contracts and customer agreements. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation's official catalog and product literature list a portable "Lithium Battery Fire Extinguisher" containing a fluorine-free water-additive surfactant liquid agent, but public business materials demonstrate that the company operates primarily as a trader and general safety equipment supplier rather than a tier-1 contractor.
- Based on primary research, the company has not disclosed further details.

Revenue & Growth in Indian Market — scored 87/100

> Secondary research could only confirm the company's general industrial safety revenue figures. Based on primary research, the company has not disclosed any specific sales revenue or growth rates for its battery fire safety products due to a lack of segmented financial disclosures or market share reports.

SUPPORTING POINTS

- Hiren Industrial Corporation reports overall annual sales revenues of approximately $1 million to $4 million within the broader fire and industrial safety sector.
- Based on primary research, the company has not disclosed further details.

Manufacturing Capacity & Localization in India — scored 85/100

> Secondary research confirms the company's localized physical presence in Mumbai and its claim of domestic fire extinguisher production. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation operates its core administrative and commercial activities out of its primary facility located in Ghatkopar East, Mumbai, Maharashtra, India.
- Product specifications from the company's catalog for its specialized water-additive/surfactant liquid Lithium Battery Fire Extinguishers explicitly state a country of origin as "Made in India".

After-Sales Service & Support Network — scored 79/100

> While basic refilling services and generalized support options are self-reported, specific quantitative metrics, such as the exact number of pan-India service centers, audited response times, or structured customer satisfaction data, could not be confirmed through secondary research, leaving the company weak or unproven on this specific operational tracking. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Hiren Industrial Corporation lists post-purchase maintenance capabilities, specifically offering refilling services for its portable and trolley-mounted fire equipment lines, as documented in the Hiren Industrial Corporation Product Catalog.
- Local commercial trade listings on platforms like Justdial highlight self-reported claims of "247 excellent service and support" alongside customer warranty options for their fire equipment installations.

Nitin Fire Protection Industries Ltd. Nitin Challengers X 71 Y 78 Overall 74

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Nitin Fire Protection Industries Ltd. provides downstream suppression kits for battery protection.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. The complete absence of verifiable evidence regarding specialized thermal runaway mitigation or containment technologies for modern advanced battery systems limits the company's standing here.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence regarding dedicated thermal runaway prevention or suppression systems for lithium-ion battery energy storage or advanced battery systems by Nitin Fire Protection Industries Ltd.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Nitin Fire Protection Industries Ltd.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Compliance with Indian & International Safety Standards — scored 72/100

> The company lists a product category for lithium-ion battery protection, but there is an absence of verifiable test records or third-party certifications regarding compliance with Indian or international battery safety standards. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Nitin Fire Protection Industries Ltd. lists "Li-ion Battery Protection" as a category under its fire suppression systems on its official catalog, indicating market entry or intent.
- Based on primary research, the company has not disclosed evidence or third-party test reports located confirming that Nitin Fire Protection Industries Ltd. possesses specific certifications for standards like AIS-156, IEC 62619, UL 1973, or UN38.3.

R&D and Innovation in Battery Fire Safety — scored 79/100

> The company includes lithium-ion battery protection within its commercial product catalog. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- The company's official product portfolio lists "Li-ion Battery Protection" under its active fire suppression system solutions.

Y — BUSINESS CAPABILITY

After-Sales Service & Support Network — scored 77/100

> The company maintains base self-reported channels for maintenance and clean-agent refilling workflows, but based on primary research, the company has not disclosed specific metrics on response times, customer satisfaction levels, and a dedicated network of operational service points across India, a gap compounded by the company's prolonged corporate restructuring under liquidation.

SUPPORTING POINTS

- Nitin Fire Protection Industries Limited provides generic technical support contact points (e.g., sales@nitinfire.com and a centralised Mumbai head office telephone line) via its updated corporate domain to handle product queries, refilling requests, and servicing bookings [1.3.1, 1.3.2].
- Publicly available corporate filings with the National Stock Exchange of India (NSE) outline that the company has been operating under an intensive, long-running Insolvency and Bankruptcy Code (IBC) liquidation process managed by a Liquidator, which culminated in a corporate restructuring sale as a going concern to a consortium led by Elysian Wealth Fund [1.4.11].

Tier-1 OEM Partnerships & Supply Contracts — scored 66/100

> Based on primary research, the company has not disclosed any contracts or partnerships with major Indian EV manufacturers, energy storage integrators, or electronics OEMs specifically for lithium-ion battery fire safety or thermal runaway prevention; the company is currently under corporate liquidation, and its public orders cover only traditional institutional and telecom fire systems.

SUPPORTING POINTS

- The company received a domestic contract from IIM Mumbai valued at Rs. 25,96,000 for the supply and installation of a comprehensive Fire Fighting System and Addressable Fire Alarm System.
- The company received a corporate purchase order from Vodafone Idea Limited for general fire detection and gas suppression systems.

Revenue & Growth in Indian Market — scored 82/100

> The company's overall Indian corporate revenue figures were verified through recent financial tracking, but based on primary research, the company has not disclosed the specific sales revenue and growth rate for battery products with fire safety features.

SUPPORTING POINTS

- Anand Rathi Financial Results reports that Nitin Fire Protection Industries Ltd. generated total annual consolidated revenues of ₹37.61 crore for FY26 and ₹10.41 crore for FY25.
- The official Nitin Fire Product Categories Portal lists "Li-ion Battery Protection" under its active product offerings, but the company's financial disclosures do not provide a segmented breakdown of sales revenue or growth rates specific to this product line.

Manufacturing Capacity & Localization in India — scored 70/100

> Secondary research confirms the locations of the company's Indian assembly units and its ongoing bankruptcy restructuring status. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Iso disclosures filed with the National Stock Exchange of India (NSE) in May 2026 verify that the company operates two local facilities in the TTC Industrial Area, Navi Mumbai (Unit 1 at Pawne MIDC and Unit 2 at Turbhe MIDC), which handle the design, assembly, testing, filling, and servicing of its gaseous fire suppression and safety systems.
- Regulatory filings from the Insolvency and Bankruptcy Board of India (IBBI) and corporate disclosures show that the company has been under liquidation since January 2022 and was recently sold as a going concern to an investor consortium, with no documented GWh-scale advanced cell/pack manufacturing capabilities or localization commitments under India’s battery Production Linked Incentive (PLI) schemes.

Regulatory Approvals & Certifications — scored 93/100

> General BIS and ISO safety frameworks for the company's suppression systems were verified, but specific certificate numbers or formal validation from advanced battery-testing authorities like ARAI or ICAT were not found in active records, leaving the evidence partial. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Nitin Fire Protection Industries Ltd. holds an ISO management system certificate (Certificate Registration No. 10001260018879, valid until May 13, 2029) issued by TUV NORD for the design, assembly, testing, filling, installation, and commissioning of its fire detection, gaseous fire suppression, and smoke detection systems.
- The company has obtained BIS certification from the Bureau of Indian Standards for its "NITIE" brand product line, ensuring compliance with national fire equipment production rules.

Exflame EXFLAME ● Challengers

X 40 Y 77 Overall 58

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 10/100

SUPPORTING POINTS

- Not available

Thermal Runaway Prevention & Suppression Systems — scored 52/100

SUPPORTING POINTS

- Standardizes a portable Clean Agent Fire Extinguisher (2 Kg, ABC Type) using HFC-236fa agent for sensitive electronic environments.
- Manufactures specialized water/foam monitors and automatic sprinkler systems to restrict bulk boundary propagation.

Battery Management System (BMS) Safety Features — scored 10/100

SUPPORTING POINTS

- Not available

Compliance with Indian & International Safety Standards — scored 86/100

SUPPORTING POINTS

- Holds formal certification across ISO 9001, ISO 14001, ISO 45001, BIS (ISI), and CE safety frameworks.
- Ensures production is executed strictly in compliance with relevant Indian standards under MSME and NSIC parameters.

R&D and Innovation in Battery Fire Safety — scored 40/100

SUPPORTING POINTS

- Utilizes an in-house, state-of-the-art testing laboratory for fire accessories and extinguishers at its Mangolpuri units.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 76/100

SUPPORTING POINTS

- Operates under the directorship of industrial veteran Mr. Rohit Gupta who brings over 25 years of firefighting industry experience.
- Maintains a strategic supply partnership with sister concern Foamtech Antifire Company for specialized foam concentrates.

Revenue & Growth in Indian Market — scored 73/100

SUPPORTING POINTS

- Grown into a major regional supplier since its incorporation in 2011, establishing a robust portfolio of industrial clients.
- Expands commercial outreach across North India from its primary operational base in the National Capital Region.

Manufacturing Capacity & Localization in India — scored 80/100

SUPPORTING POINTS

- Operates two dedicated domestic production hubs, Unit-1 and Unit-2, situated in the Mangolpuri Industrial Area, New Delhi.
- Fully localizes fabrication of landing valves, fire hoses, fire doors (MS/SS), and customizable fire hose cabinets.

After-Sales Service & Support Network — scored 72/100

SUPPORTING POINTS

- Offers tailored client support and custom product configurations built exactly to client technical specifications.
- Leverages a cross-regional sales and services network through its registered corporate office at Mangolpuri, Delhi.

Regulatory Approvals & Certifications — scored 84/100

SUPPORTING POINTS

- Fully certified under ISO 9001, ISO 14001, and ISO 45001 international management standards.
- Holds Bureau of Indian Standards (BIS/ISI) and CE marks across its full range of firefighting equipment and accessories.

Foamtech Antifire Company Foamtech ● Trailblazers X 86 Y 83 Overall 84

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 68/100

> Foamtech Antifire Company is strictly a firefighting suppression system and chemical agent manufacturer. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Foamtech Antifire Company promotes its "Lith Cool Agent" on Foamtech Antifire Company Instagram, which is an external fire suppression agent formulated to cool lithium-based fires and suppress thermal runaway rather than a cell chemistry or material.
- The company manufactures Ternary Eutectic Chloride (TEC) base dry chemical powder under its Flamax line for lithium metal fires, as detailed in its product documentation on Foamtech Antifire Product Catalog.

Thermal Runaway Prevention & Suppression Systems — scored 93/100

> Secondary research confirms that the company manufactures commercial fire suppression agents and portable extinguishers designed specifically to suppress lithium-ion battery fires and mitigate thermal runaway, but based on primary research, the company has not disclosed third-party verified test reports or independent data showing successful test outcomes, leaving the evidence self-reported and only partially commercialized.

SUPPORTING POINTS

- The company manufactures a specialized water additive called Foamer Lith-Cool Agent and model CA/2026 9 kg encapsulator agent-based battery fire extinguisher engineered to rapidly absorb heat, drop surface temperatures, and interrupt the chain reaction of thermal runaway in lithium-ion batteries, as outlined on the Foamtech Antifire Foam Concentrates Catalog.
- The company produces the Flamax Class D Fire Extinguishing Agent, which fuses above 650°C to form a solid crust that isolates oxygen and halts heat radiation on active metal fires including lithium, according to the Foamtech Dry Chemical Powder Product Page.

Battery Management System (BMS) Safety Features — scored 100/100

> Foamtech mentions basic system-level interfacing with external BMS panels and generic cell-level monitoring concepts in promotional updates. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Social media marketing assets briefly detail an integrated "cell-level BMS monitoring" framework meant to detect out-of-range temperature or voltage and isolate modules.
- Corporate technical updates note basic compatibility and interfacing options between their physical suppression units and external "Intelligent Battery Management Systems (BMS)" or Building Management Systems (BMS) to process automated fire alarms.

Compliance with Indian & International Safety Standards — scored 78/100

> While general industrial fire and chemical safety certifications such as UL, BIS, and ISO were verified, public evidence or third-party test reports confirming compliance with explicit battery-specific safety standards like AIS-156, IEC 62619, UL 1973, or UN38.3 are absent, leaving the company unproven on this specific measure.

SUPPORTING POINTS

- Foamtech Antifire Company's in-house facility FT Test Lab is accredited by the National Accreditation Board for Testing and Calibration Laboratories (NABL) under ISO/IEC 17025:2017 (Certificate TC-13311).
- The company's general fire protection product range, including its AFFF 3% foam concentrate, holds official UL-162 listing from Underwriters Laboratories alongside Bureau of Indian Standards (BIS/ISI) and CE certifications.

R&D and Innovation in Battery Fire Safety — scored 93/100

> Secondary research confirms that the company actively commercializes water-additive agents and total flooding systems for lithium-ion applications, but based on primary research, the company has not disclosed independent R&D expenditures, co-authored academic research, or proprietary patents, leaving the evidence partial.

SUPPORTING POINTS

- The company commercialized its proprietary Foamer Lith Cool Aqueous Water Additive Solution, an advanced cooling concentrate engineered to control thermal runaway events in lithium-ion batteries.
- The company provides custom-engineered automatic fire suppression systems utilizing an integrated testing facility.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 76/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Foamtech Antifire Company manufactures a specialized aqueous solution for lithium-ion fires called FOAMER LITH-COOL.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Revenue & Growth in Indian Market — scored 93/100

> The company has a verifiable product line for lithium-ion battery fire suppression, but its market share, growth metrics, and revenue specifically within this segment are completely unsegregated and lack public third-party verification. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Foamtech Antifire Company offers a dedicated specialized product called the Foamer Lith Cool Aqueous Water Additive Solution designed to reduce surface temperature and control thermal runaway in electric vehicle battery facilities and storage warehouses.
- Financial profiles indicate that Foamtech Antifire Private Limited operates with an overall revenue range of ₹10–₹50 Cr ($4 million) for its entire portfolio of conventional and specialized fire protection systems. Based on primary research, the company has not disclosed granular breakdown or isolated sales figures.

Manufacturing Capacity & Localization in India — scored 65/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Foamtech Antifire Company operates manufacturing units in Haridwar, Uttarakhand, and Kundli, Sonipat, Haryana, producing conventional fire suppression materials and systems.
- Based on primary research, the company has not disclosed evidence demonstrating localized lithium-ion cell or battery pack assembly capacity, GWh metrics, or PLI scheme commitments relevant to battery manufacturing.

After-Sales Service & Support Network — scored 89/100

> The company provides broad, self-reported statements on maintenance support but lacks third-party verified service center volumes or structured after-sales network data for advanced battery applications. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Foamtech Antifire Company offers generalized installation and maintenance support for its core fire suppression systems.
- The company maintains its corporate headquarters in Delhi and a manufacturing/refilling facility in Sonipat, Haryana. Based on primary research, the company has not disclosed a widespread after-sales service network or warranty frameworks specifically dedicated to its battery safety line.

Regulatory Approvals & Certifications — scored 92/100

> The company shows a clear manufacturing capability for lithium-ion fire suppression materials, but its compliance claims in this niche are self-reported and lack verified third-party battery-specific certifications or certificate numbers from the designated automotive or regulatory bodies.

SUPPORTING POINTS

- Foamtech Antifire Company manufactures a specialized aqueous water additive solution called "FOAMER LITH-COOL AGENT" specifically designed to combat lithium-ion battery fires and suppress thermal runaway.
- While Foamtech holds extensive certifications for its broader industrial firefighting powders and foam concentrates, it lists no specific third-party regulatory approvals or certificate numbers from Indian agencies like ARAI or ICAT explicitly validating its LITH-COOL agent for battery safety systems.

Ionex Fire IonexFire ● Trailblazers X 83 Y 80 Overall 82

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 70/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Ionex Fire focuses exclusively on external fire suppression solutions, such as encapsulator and vermiculite-based fire extinguishers, rather than developing fire-resistant battery chemistries or cell and pack materials.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 93/100

> The company shows self-reported and partial evidence of portable suppression capabilities for smaller applications rather than third-party verified large-scale integration.

SUPPORTING POINTS

- Ionex Fire manufactures specialized extinguishers utilizing agents like TerraSafe TS-3E and vermiculite dispersions intended to manage lithium-ion battery thermal runaway.
- Internal specifications indicate the Ionex-EA solution addresses small-scale consumer battery fires up to 600 Wh, while lacking formal standardized industrial certifications.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Compliance with Indian & International Safety Standards — scored 87/100

> Ionex Fire provides partial, self-reported evidence of compliance through its DOT transport compliance and use of a UL-certified raw material agent, but it lacks direct third-party certifications against key battery safety benchmarks like AIS-156, IEC 62619, or UL 1973.

SUPPORTING POINTS

- Ionex Fire (operating under Marsol Technologies) states that its Ionex-EFX extinguisher utilizes a UL-certified agent (aqueous dispersion of vermiculite) and complies with DOT transport and safety requirements.
- The company reports that its Ionex-EA model is not certified under a standardized lithium-ion fire rating system due to a lack of defined industry criteria, but adheres to DOT standards and relies on internal testing up to 600 Wh.

R&D and Innovation in Battery Fire Safety — scored 93/100

> Ionex Fire shows a clear, verifiable product capability and specialized material application for battery fires, but its R&D framework relies largely on internal validation and integrated partner materials rather than an extensive, market-leading proprietary patent suite or verified external research.

SUPPORTING POINTS

- Marsol Technologies launched the Ionex Fire Extinguisher line (including Ionex-EA and Ionex-EFX), which utilizes advanced cooling chemistry like the TerraSafe TS-3E encapsulator agent and UL-certified aqueous dispersions of vermiculite specifically optimized to suppress lithium-ion battery fires.
- Product documentation notes internal testing and performance validation showing that the technology is engineered to successfully control and suppress thermal runaway in battery packs up to 600 Wh.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 70/100

> Secondary research looked for contracts, named customers, transaction values, and partnership durations with Indian electric vehicle manufacturers or energy storage OEMs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Ionex Fire showcases its product portfolio—including the Ionex-EA and Ionex-EFX battery fire extinguishers—entirely as general commercial safety products for personal mobility, electrical equipment, and electronics, with no mention of secured tier-1 OEM clients.
- Public distribution records and corporate announcements from its partner platform reveal product promotional presence at trade exhibitions like Intersec 2025, but fail to disclose any contract values, duration of agreements, or specific supply partnerships with major electric vehicle manufacturers or energy storage integrators.

Revenue & Growth in Indian Market — scored 87/100

> The company's local manufacturing presence in Gurgaon and international launch events are verifiable, but specific domestic sales figures, financial growth rates, and market share statistics are entirely absent from public financial reports or third-party market data.

SUPPORTING POINTS

- The company launched its product line internationally at the Intersec Exhibition 2025 in collaboration with Marsol Technologies Inc, showcasing its specialized lithium-ion battery fire suppression systems to tap into global distribution networks.
- The brand actively operates from its manufacturing and corporate presence at Surat Nagar, Industrial Area, Gurgaon, Haryana, showcasing local infrastructure to cater to the domestic market.

Manufacturing Capacity & Localization in India — scored 75/100

> Secondary research could confirm only the company's local industrial site address in Gurgaon. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Ionex Fire lists its operational address at 4652/279, Surat Nagar, Industrial Area, Gurgaon, Haryana-122001, indicating a local corporate and operational hub in India.
- Public specifications for its Ionex-EA and Ionex-EFX battery fire extinguishers mention the utilization of eco-friendly encapsulator and aqueous dispersion of vermiculite agents tested for lithium-ion fires up to 600 Wh, but do not outline local industrial battery factory locations, GWh output capacity, or cell/pack production line details.

After-Sales Service & Support Network — scored 77/100

> While direct manufacturer contact channels and basic support inquiry mechanisms are active, an extensive nationwide network of service centers, formal warranty structures, specific response times, or customer satisfaction metrics across India could not be confirmed through secondary research, leaving the company with unproven, weak evidence here. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Ionex Fire provides corporate contact metrics and a direct service inquiry channel managed from its main facility located at 4652/279, Surat Nagar, Industrial Area, Gurgaon, Haryana, via its official Ionex Fire Contact Portal.
- Product documentation via distribution partner Marsol Technologies includes technical specification forms and a "Request More Info" mechanism to coordinate direct product support and maintenance instructions for their Ionex-EA systems.

Regulatory Approvals & Certifications — scored 90/100

> International agent-level compliances (UL and DOT) are verified. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Ionex-EFX utilizes a UL-certified agent based on an aqueous dispersion of vermiculite to suppress lithium-ion battery fires, as reported on the Ionex Fire Product Page.
- Ionex-EA and Ionex-EFX firefighting agents are fully compliant with Department of Transportation (DOT) requirements for transport and usage safety, as documented by distribution partner Marsol Technologies.

CaloryTech ● Trailblazers

X 87 Y 78 Overall 82

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 94/100

> The company's proprietary GrafX flame-retardant pack materials and internal safety test results were verified, but its use of inherently safer cell chemistries and formal independent third-party test certificates could not be conclusively validated beyond self-reported data, leaving the evidence partial. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- CaloryTech developed GrafX, a proprietary passive thermal management material composed of a fire-resistant graphite and heat-absorbing phase-change material composite designed for integration into lithium-ion battery pack configurations.
- In a company case study testing a 12V 15Ah battery pack, the implementation of CaloryTech's GrafX composite successfully mitigated thermal runaway propagation, delaying the onset to over 45 minutes and enabling the system to meet Indian AIS-156 Phase 2 safety standards.

Thermal Runaway Prevention & Suppression Systems — scored 91/100

> Secondary research confirms CaloryTech's development of passive phase-change thermal management materials and self-reported compliance metrics, but based on primary research, the company has not disclosed external third-party test reports or extensive commercial client deployments, keeping the evidence self-reported.

SUPPORTING POINTS

- Comparative thermal abuse testing of a 12V, 15Ah lithium-ion battery pack demonstrated that while an unprotected pack experienced violent explosion in under 10 minutes, integration of CaloryTech's GrafX (a graphite-phase change material composite) successfully prevented thermal runaway propagation and delayed safety onset past 45 minutes.
- CaloryTech's plug-and-play GrafX passive cooling system uses proprietary CTI GrafX Technology to rapidly dissipate heat across battery pack configurations, enabling electric vehicles to satisfy India's mandatory AIS 156 Phase 2 battery fire safety compliance standards.

Battery Management System (BMS) Safety Features — scored 75/100

> Real-world thermal propagation compliance via its GrafX technology passing AIS 156 Phase 2 certification was verified, but based on primary research, the company has not disclosed proprietary BMS monitoring algorithms, fail-safe mechanisms, or sensor response metrics, since the company focuses strictly on material-level and physical thermal management solutions.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed information for this parameter. 2.2, 1.3.1].
- Public technical communications by the company focus entirely on physical thermal runaway prevention and mitigating "inaccurate BMS readings" caused by poor thermal regulation, treating the BMS as a separate hardware component altogether [1.2.6].

Compliance with Indian & International Safety Standards — scored 87/100

> Secondary research confirms self-reported claims and internal test metrics demonstrating compliance with regional Indian AIS-156 rules. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- An internal test report shared by founder Siddique Khateeb for a 12V, 15Ah lithium-ion battery pack using CaloryTech's GrafX passive thermal management system demonstrated that the system delayed thermal runaway onset to over 45 minutes with zero propagation, exceeding the mandates of India's AIS-156 Phase-2 standards.
- Product showcases and promotional documentation presented by Afzaal Mohammed (Head of Finance & Business Development at CaloryTech) state that their GrafX graphite-PCM composite system enables battery packs to pass AIS-156 (Amendment 3) certification tests by rapidly dissipating heat.

R&D and Innovation in Battery Fire Safety — scored 89/100

> Secondary research confirms the company's proprietary materials innovation and patented technology claims, but based on primary research, the company has not disclosed formal academic research institution collaborations or public R&D expenditure data, likely reflecting its status as a small, unfunded private startup, leaving the evidence self-reported and partial.

SUPPORTING POINTS

- Empirical testing and development of its proprietary GrafX technology, an optimized Graphite-Phase Change Material (PCM) composite that completely mitigated thermal runaway propagation and delayed its onset to over 45 minutes in a 12V 15Ah lithium-ion battery pack.
- Founder Siddique Khateeb was selected to participate in the Fall 2024 Landis Family HardTech fellowship program at the mHUB innovation center to advance the development of these novel battery thermal management materials.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 76/100

> The company claims capability to pass Indian EV safety standards with its GrafX product. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- CaloryTech's internal testing of its GrafX thermal management system indicates capability to exceed AIS-156 Phase 2 standards to prevent 2- and 3-wheeler EV battery fires, but no external client contracts are mentioned.

Revenue & Growth in Indian Market — scored 85/100

> CaloryTech is an early-stage, unfunded Indian startup that has developed its "GrafX" thermal management system for battery fire prevention, but its operations remain small-scale, with estimated revenue under ₹10 Cr, given a lack of verified commercial scaling or market share documentation.

SUPPORTING POINTS

- Tracxn reports that CaloryTech Innovation Private Limited generated less than ₹10 Cr ($1.2 million USD) in revenue for the financial year ending March 31, 2025.
- Prospeo estimates CaloryTech's annual revenue at $342,220.

Manufacturing Capacity & Localization in India — scored 72/100

> The company is a micro-enterprise based in Bangalore that focuses on providing customized component/material samples for pack integration rather than possessing giga-scale cell or pack manufacturing infrastructure. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- CaloryTech operates out of its headquarters in Bengaluru, India, providing graphite composite phase change materials (GrafX) for lithium-ion battery pack thermal management.
- Based on primary research, the company has not disclosed evidence regarding its exact factory capacity figures in GWh or any official localization commitments or approvals under India's national PLI schemes.

After-Sales Service & Support Network — scored 65/100

> The company offers standard direct communication channels for initial engineering support, but it lacks any documented after-sales service centers or verifiable support infrastructure across India. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- CaloryTech provides direct technical contact via email and a dedicated mobile number to assist customers with custom-made samples and technical implementation of their GrafX technology.
- Based on primary research, the company has not disclosed evidence regarding a formalized after-sales support network, physical service points, response times, or customer satisfaction metrics in India.

Regulatory Approvals & Certifications — scored 90/100

> The company makes explicit self-reported claims regarding compliance and its product's ability to clear AIS 156 Phase 2 tests, but no independent third-party certificate numbers or direct regulatory approval filings from ARAI, ICAT, or BIS were found in public records. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- CaloryTech states that its thermal management system (TMS) product, GrafX, is designed to rapidly dissipate heat and prevent thermal runaway propagation, allowing battery packs to easily pass the AIS 156 Phase 2 certification tests.

Varistor Technologies Pvt. Ltd. VariEx ● Trailblazers X 79 Y 77 Overall 78

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 70/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Thermal Runaway Prevention & Suppression Systems — scored 89/100

> Secondary research confirms the availability of active and passive suppression agents like aerosol systems intended for battery environments. Based on primary research, the company has not disclosed any formal third-party test outcomes or wide-scale integration into major commercial battery packs.

SUPPORTING POINTS

- Varistor Technologies Pvt. Ltd., through its premier brand VariEx, markets an Automatic Aerosol Fire Suppression Device designed to automatically detect and suppress fires at early stages inside enclosed electrical compartments and battery cabinets.
- The company promotes self-activating suppression products via its corporate media specifically targeting EV charging safety and mitigation of thermal runaway events.

Battery Management System (BMS) Safety Features — scored 73/100

> Based on primary research, the company has not disclosed verifiable evidence of active electronic monitoring, sensor response times, or preventative BMS logic for this manufacturer, since it operates exclusively as a mechanical and chemical fire suppression provider.

SUPPORTING POINTS

- Varistor Technologies Pvt. Ltd., operating through its VariEx® brand, focuses on downstream physical protection and automatic clean agent or aerosol fire suppression systems for electrical and battery environments without proprietary active BMS electronics development.
- Corporate product catalogs and technical data indicate that the company's installations rely on mechanical direct low-pressure tubings and gaseous expellants rather than digital fail-safe algorithms or cell-balancing microcontrollers.

Compliance with Indian & International Safety Standards — scored 81/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- As documented on VariEx Products Certification, Varistor Technologies Pvt. Ltd. holds standard industrial certifications such as ISI marking for traditional fire extinguisher cylinders and basic ISO compliance.
- According to product listings on VariEx Online Portal, the company offers automatic tube-based and self-activating aerosol fire suppression systems designed for generic enclosed spaces and electrical panels, but lacks any published third-party test reports or homologation files tying these systems specifically to battery standards.

R&D and Innovation in Battery Fire Safety — scored 82/100

> While standard electrical suppression manufacturing capabilities were verified, concrete evidence regarding specific battery fire safety patents, R&D expenditures, or academic research collaborations remains unproven.

SUPPORTING POINTS

- VariEx manufactures the VariEx Clean Agent Gas DLP Fire Suppression System and automatic gas-based setups for sensitive electrical environments.
- The company maintains internal operations where its VariEx division functions as a development wing focused on system integration and safety engineering.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 68/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Revenue & Growth in Indian Market — scored 85/100

> The company has visible general corporate financials under ₹10 Cr, but lacks third-party verified breakdown or specific reported segments for advanced battery fire safety or thermal runaway suppression revenue in India, indicating weak or unproven capability strictly on this specialized measure. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Financial data lists the general corporate revenue of Varistor Technologies Private Limited at ₹4.58 Cr for the financial year ending March 31, 2025, but does not segment any dedicated growth or specific revenue for battery fire safety systems.
- While the company's brand "VariEx" is known for general fire extinguishers and safety solutions, it is mentioned in secondary industrial market listings alongside armored vehicle fire suppression systems without publicly isolated sales figures for advanced battery fire applications.

Manufacturing Capacity & Localization in India — scored 72/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Varistor Technologies Pvt. Ltd. operates primarily as a commercial supplier, dealer, and system installer under brands such as VariEx and SolarClue. Based on primary research, the company has not disclosed further details.

After-Sales Service & Support Network — scored 77/100

> The company shows proof of standard maintenance operations, but based on primary research, the company has not disclosed an after-sales network tailored for advanced battery safety systems.

SUPPORTING POINTS

- Varistor Technologies Pvt. Ltd. operates a fire safety vertical under the brand VariEx, which supplies fire extinguishers and suppression systems, though public evidence regarding specialized technical support networks or response metrics for battery systems is absent.
- The company indicates general post-sales capability through localized hiring and generalized support claims, but lacks third-party validated customer satisfaction metrics or a verified nationwide service footprint.

Regulatory Approvals & Certifications — scored 82/100

> Varistor Technologies manufactures and distributes general fire safety equipment with standard ISI/CE marks under its VariEx brand, but it lacks third-party verified, specialized regulatory approvals such as ARAI or ICAT that are tailored specifically for advanced battery fire protection and thermal runaway suppression. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- VariEx® (the fire safety brand of Varistor Technologies Pvt. Ltd.) certifies broadly that its generic line of fire safety systems—including extinguishers and basic suppression products—carries ISI markings and CE approvals.
- Based on primary research, the company has not disclosed information for this parameter.

Varia FireTech Industries V-Safe ● Trailblazers X 53 Y 77 Overall 65

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 45/100

SUPPORTING POINTS

- Formulates a Ternary Eutectic Chloride (TEC) based extinguishing agent specifically designed for treating lithium and its alloys used in batteries.

Thermal Runaway Prevention & Suppression Systems — scored 75/100

SUPPORTING POINTS

- Manufactures the V-Safe portable and wheeled Class D fire extinguishers featuring custom diffusers that optimize soft flow for lithium-ion battery fires.
- Designs custom fire suppression tubing and gas total flooding fire systems tailored to enclosed user hazard requirements.

Battery Management System (BMS) Safety Features — scored 10/100

SUPPORTING POINTS

- Not available

Compliance with Indian & International Safety Standards — scored 85/100

SUPPORTING POINTS

- Products pass exhaustive testing to meet the BIS and IS 15683:2018 standard for portable fire extinguishers in India.
- Fire suppression components incorporate CE approved brass valves equipped with non-return valves (NRV) and safety releases.

R&D and Innovation in Battery Fire Safety — scored 50/100

SUPPORTING POINTS

- Operates a 70,000 square foot modern manufacturing infrastructure and testing facility in Jamnagar, India.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 78/100

SUPPORTING POINTS

- Supplies Class D Ternary Eutectic Chloride (TEC) based extinguishers applicable on lithium battery fires to battery manufacturing units.
- Partners with Universal Fire and Safety Services Limited for public tenders and contracts.

Revenue & Growth in Indian Market — scored 72/100

SUPPORTING POINTS

- Secured standard fire extinguisher and battery supply orders through state and public sector bids such as IOCL-PLHO.
- Serves multiple local and regional markets with prominent industrial distributors across Gujarat including Ahmedabad.

Manufacturing Capacity & Localization in India — scored 82/100

SUPPORTING POINTS

- Operates a modern manufacturing facility located in Jamnagar, Gujarat spread across 70,000 square feet.
- Fabricates extinguisher bodies locally using CR 4 graded sheets with internal and external pure polyester coatings.

After-Sales Service & Support Network — scored 70/100

SUPPORTING POINTS

- Offers direct operational consultations and maintenance contact support through its corporate office at Hapa Industrial Estate, Jamnagar.
- Manages multi-regional distribution and support channels near major industrial centers like Kheda and Surat.

Regulatory Approvals & Certifications — scored 85/100

SUPPORTING POINTS

- Fully certified under the Bureau of Indian Standards conforming to the IS 15683:2018 product standard.
- Uses valves with European CE approval equipped with non-return valves (NRV) and safety releases.

Sheetla Techno Industries STI ● Evolving Players X 76 Y 74 Overall 75

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 67/100

> Based on primary research, the company has not disclosed any utilization of advanced battery chemistries, flame-retardant pack-level materials, or cell safety test results, since the company operates exclusively as an external sheet metal fabricator, leaving no evidence here.

SUPPORTING POINTS

- Sheetla Techno Industries manufactures external sheet metal battery cabinets, EV battery enclosures, and heavy-duty battery housing solutions from mild steel or stainless steel (SS 304 & SS 316) rather than cell-level chemistries.
- Based on primary research, the company has not disclosed information for this parameter.

Thermal Runaway Prevention & Suppression Systems — scored 87/100

> Secondary research confirms that the company offers passive structural containment through metal battery cabinets and enclosures, but based on primary research, the company has not disclosed any evidence of active cooling, venting, built-in fire suppression agents, or third-party verified fire propagation test outcomes, leaving the evidence partial and self-reported.

SUPPORTING POINTS

- Sheetla Techno Industries (STI Brand) manufactures customized sheet metal battery cabinets, EV battery enclosures, and heavy-duty lithium battery storage boxes engineered to provide basic thermal protection and structural containment for electric vehicles across India.
- Product specifications from the company's official corporate catalog on 13SQFT emphasize its manufacturing focus on passive powder-coated mild steel, structural steel, and stainless steel fabrication rather than active thermal management or integrated fire suppression systems.

Battery Management System (BMS) Safety Features — scored 68/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Secondary research attempted to verify active real-time monitoring, fault detection algorithms, and electronic fail-safes, but could verify none of these features, since the company's capabilities are limited strictly to passive structural steel fabrication, metal battery cabinets, and physical safety enclosures.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed information for this parameter.
- Based on primary research, the company has not disclosed information for this parameter.

Compliance with Indian & International Safety Standards — scored 81/100

> Sheetla Techno Industries is a verified physical manufacturer of custom sheet metal battery boxes and cabinets in Gurugram. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to the product portfolio published by Sheetla Techno Industries, the company utilizes ISO-standard sheet metal fabrication processes to engineer custom heavy-duty EV battery enclosures and metal containment cabinets.
- The company self-reports adhering to nationwide national and international generic quality norms and necessary manufacturing industry standards across its broader industrial storage and furniture lines.

R&D and Innovation in Battery Fire Safety — scored 79/100

> Secondary research confirms that the company manufactures customized battery enclosures marketed for thermal safety, but based on primary research, the company has not disclosed any proprietary R&D programs, patents, or institutional partnerships for active battery fire safety innovation.

SUPPORTING POINTS

- Sheetla Techno Industries fabricates powder-coated sheet metal enclosures and EV lithium battery storage cabinets designed for two-, three-, and four-wheeler electric vehicles, self-reporting that they are engineered for thermal protection and safety on the Sheetla Techno Industries Product Portal.
- Based on primary research, the company has not disclosed information for this parameter.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 76/100

> The company manufactures EV battery enclosures, but the lack of verifiable Tier-1 OEM partnerships limits its standing here. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sheetla Techno Industries manufactures structural sheet metal enclosures and metal battery cabinets targeted for electric vehicles across India.
- Based on primary research, the company has not disclosed evidence regarding signed supply contracts or official partnerships with named Tier-1 automotive OEMs or energy storage integrators.

Revenue & Growth in Indian Market — scored 78/100

> The company manufactures passive sheet metal enclosures for EV batteries. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sheetla Techno Industries (STI Brand) is verified as a manufacturer of material handling equipment, sheet metal storage cabinets, and basic EV battery metal enclosures/cabinets.
- Based on primary research, the company has not disclosed evidence regarding financial statements, market share data, or granular sales revenue specifically tied to battery products with proactive fire safety features or thermal runaway prevention systems.

Manufacturing Capacity & Localization in India — scored 76/100

> The company manufactures foundational mechanical metal enclosures and cabinets rather than active battery cells or complex pack assemblies, and it lacks verifiable metrics for battery cell GWh manufacturing capacity or PLI alignment. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sheetla Techno Industries manufactures passive safety metal battery cabinets, EV battery metal enclosures, EV battery boxes, and powder-coated lithium battery storage cabinets at its facility in IMT Manesar, Gurugram, Haryana.
- Based on primary research, the company has not disclosed evidence regarding advanced cell manufacturing, pack assembly operations, Gigawatt-hour (GWh) capacity metrics, or localization commitments under the Production Linked Incentive (PLI) scheme.

After-Sales Service & Support Network — scored 67/100

> The company lacks third-party verified data, concrete metrics, or dedicated nationwide infrastructure detailing response times and service networks for specialized battery fire safety applications, relying instead on generic self-reported post-purchase guidelines and remote support. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Sheetla Techno Industries (STI Brand) lists structural features for its EV metal battery cabinets—such as fire-resistant structures and ventilation support—but its published after-sales support consists of generalized text regarding "post-purchase support" and "warranty options" without verifiable metrics.
- E-commerce warranty data for standard STI hardware products outlines a highly limited 1-Month Domestic Warranty, specifying that technical support is executed via localized WhatsApp video calls rather than a formal on-site service center network across India.

Regulatory Approvals & Certifications — scored 72/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Sheetla Techno Industries manufactures fire-resistant metal battery cabinets and EV enclosures.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Nest Protection ● Evolving Players

X 75 Y 73 Overall 74

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 65/100

> Secondary research confirms that the company focuses strictly on external active fire hardware. Based on primary research, the company has not disclosed any use or development of safer battery chemistries (like LFP or solid-state) or internal flame-retardant pack materials, since Nest Protection is an equipment integrator rather than a battery pack manufacturer.

SUPPORTING POINTS

- According to primary research, a commercial product listing confirms that Nest Protection manufactures pre-engineered "Fire Detection Tube Suppression Systems" utilizing active clean agents (FK-5-1-12/Novec 1230) aimed at external battery bank protection rather than internal cell engineering.
- The company's official business profile on ExportersIndia explicitly details their capabilities as a supplier and contractor for fire extinguishers, gas detectors, and control panels, showing no presence in materials science or chemical engineering for battery cells.

Thermal Runaway Prevention & Suppression Systems — scored 75/100

> While general suppression listings are present, specific thermal runaway test outcomes or integration into battery energy storage systems could not be confirmed through secondary research due to a lack of specialized evidence. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, Nest Protection is listed as a supplier and system integrator offering Novec 1230 Suppression Systems and Fire Detection Tube Suppression Systems on its commercial business profile.
- The company provides total flood type delivery platforms designed to protect server rooms and control spaces as detailed in its Product Catalog Documentation.

Battery Management System (BMS) Safety Features — scored 78/100

> The verified product inventory and service lines focus entirely on downstream mechanical gas detection, containment, and suppression hardware. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, product profiles show that Nest Protection supplies pre-engineered direct/indirect low-pressure tube fire suppression systems utilizing FK-5-1-12/Novec 1230 agents for battery rooms, server rooms, and electrical panels.
- Corporate credentials from Nest Protection's Official Site indicate their core technical scope focuses on integrating hardware-based addressable or conventional fire alarm panels (IS 2189 compliant) with third-party automation systems and localized mechanical tube containment networks.

Compliance with Indian & International Safety Standards — scored 78/100

> Secondary research confirms baseline material certifications for the system's component sub-assembly (UL and CE tube markings). Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Product specifications listed on Aajjo Product Catalogue confirm that Nest Protection's pre-engineered systems employ a polyamide fire detection tube that holds UL and CE approval markings for baseline quality compliance.
- According to primary research, trade and manufacturing profiles hosted on a supplier directory indicate the company functions primarily as a local regional manufacturer of industrial clean agent systems, but lack documented independent type-test validations specifically contextualised to global battery testing registries.

R&D and Innovation in Battery Fire Safety — scored 80/100

> Secondary research confirms the availability of the company's clean-agent tube-based suppression systems configured for server and battery rooms. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, Nest Protection markets a pre-engineered Fire Detection Tube Suppression System listed on its commercial catalog product page which utilizes FK-5-1-12 (Novec 1230) clean agent for panel-level suppression, noting application areas such as battery rooms.
- The company acts as a regional supplier, installer, and systems integrator sourcing from established third-party brands rather than conducting original R&D, as documented on the Nest Protection Official Website where they detail implementing setups from brands like Ravel, Bosch, Honeywell, Agni, Cooper, and Notifier.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 66/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Nest Protection operates primarily as an industrial fire safety contractor and general system integrator in India, with no publicly disclosed strategic tier-1 OEM supply contracts or partnerships specific to the advanced battery fire safety or electric vehicle manufacturing industries.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Revenue & Growth in Indian Market — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Manufacturing Capacity & Localization in India — scored 68/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence indicating that the company Nest Protection produces technologies, materials, or systems designed to prevent, detect, contain, or suppress thermal runaway events specifically for advanced battery systems or possesses GWh manufacturing capacity under PLI schemes.
- Public commercial listings indicate Nest Protection is a Gurugram-based manufacturer of traditional digitally addressable fire alarm systems, wireless smoke detectors, and standard fire extinguishers for commercial and industrial facilities rather than advanced battery cell or pack thermal safety assembly.

After-Sales Service & Support Network — scored 100/100

> Nest Protection has demonstrated clear verifiable capability for after-sales support across India through regional offices and a 24/7 hotline.

SUPPORTING POINTS

- Nest Protection provides pan-India operations with office presences listed in Mumbai, Delhi, Bengaluru, and Chennai, featuring over 50 certified engineers and a dedicated 24/7 emergency support line for breakdown maintenance and Annual Maintenance Contracts.
- According to primary research, customer satisfaction metrics indicate a self-reported rating of 4.6/5 based on 33 reviews on its business listing, alongside a 77% customer response rate.

Regulatory Approvals & Certifications — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed the industrial fire systems contractor Nest Protection holding battery-specific fire safety certifications or approvals from regulatory bodies like ARAI, ICAT, or BIS for advanced battery systems.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

Daasnav Solutions Private Limited Daasnav ● Evolving Players X 54 Y 85 Overall 70

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 10/100

SUPPORTING POINTS

- Not available

Thermal Runaway Prevention & Suppression Systems — scored 94/100

SUPPORTING POINTS

- Engineered the specialized model DS-PF-F002-Li-Ion 2-Liter and 6-Liter Lithium-Ion Battery Fire Extinguishers.
- Supplies specialized Fire Burst Tubes (suppression tube systems) targeted at localized thermal runaway containment.
- Manufactures high-pressure water mist and Compressed Air Foam Systems (CAFS) configured for complex industrial fire risks.

Battery Management System (BMS) Safety Features — scored 10/100

SUPPORTING POINTS

- Not available

Compliance with Indian & International Safety Standards — scored 80/100

SUPPORTING POINTS

- Qualified as a vetted L1 supplier under public procurement evaluations by the Ministry of Defence and Bharat Electronics Limited (BEL) for Li-ion extinguishers.

R&D and Innovation in Battery Fire Safety — scored 78/100

SUPPORTING POINTS

- Operates a proprietary fire testing ground authorized to evaluate complex, multi-tier fire rating parameters.
- Equips its Sector 65, Noida hub with high-end machinery and dedicated laboratories for quality analysis.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 88/100

SUPPORTING POINTS

- Key technology supplier to elite tier public enterprises and PSUs including ONGC, GAIL, IOCL, and NTPC units.
- Partnered with central government departments including the Airport Authority of India, Indian Railways, and NDRF.

Revenue & Growth in Indian Market — scored 84/100

SUPPORTING POINTS

- Achieved a substantial annual corporate turnover of ₹47.6 Crores since its founding in 2020.
- Secured recent high-value commercial bid awards including a ₹15.0 Lakhs HPCL contract and a ₹24.7 Lakhs North Central Railway order.

Manufacturing Capacity & Localization in India — scored 86/100

SUPPORTING POINTS

- Operates a state-of-the-art manufacturing plant at B-135, Sector 65, Noida, Uttar Pradesh.
- Features proprietary domestic production of Carbon Composite Fire Extinguishers with HDPE/PET liners and Aramid/Kevlar fiber weaves.

After-Sales Service & Support Network — scored 80/100

SUPPORTING POINTS

- Deploys expert certified technicians providing automated inspection, equipment installations, and system monitoring.
- Maintains an active centralized helpdesk via a national toll-free support line (18004198303).

Regulatory Approvals & Certifications — scored 89/100

SUPPORTING POINTS

- High pressure water mist and compressed air foam systems (CAFS) carry official PESO approvals with a 20-year shelf life.
- Products carry official ISI marks conforming to standard IS 15683 along with formal MSME and ISO certifications.

Nasa Fire Technology Limited NASA Fire ● Evolving Players X 60 Y 64 Overall 62

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 45/100

SUPPORTING POINTS

- Developed specialized flame-retardant chemical granules for tailored industrial safety and material applications.

Thermal Runaway Prevention & Suppression Systems — scored 85/100

SUPPORTING POINTS

- Invented independent automatic fire detecting and extinguishing units operating purely on gravity without external energy or battery backup.
- Features proprietary maintenance-free leak-proof valve assemblies and stainless steel 304 devices for early-stage fire suppression.

Battery Management System (BMS) Safety Features — scored 20/100

SUPPORTING POINTS

- Systems function autonomously without relying on external power or internal electrical circuits.

Compliance with Indian & International Safety Standards — scored 80/100

SUPPORTING POINTS

- Operational processes and product capabilities are validated under TUV reliable operation standards.
- Manufacturing processes conform to ISO 9001 quality management guidelines.

R&D and Innovation in Battery Fire Safety — scored 70/100

SUPPORTING POINTS

- Dedicated over 38 years to intensive research analyzing operational conditions and human psychology to innovate first-aid fire safety.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 68/100

SUPPORTING POINTS

- Manufactures and supplies advanced Linear Heat Sensing Tubing Based Clean Agent Suppression Systems for industrial equipment enclosures.
- Designs specialized flame-retardant chemical granules and proprietary gas mixer-based extinguishers for industrial safety integrators.

Revenue & Growth in Indian Market — scored 62/100

SUPPORTING POINTS

- Maintained continuous, active operations in the domestic market since its initial commercial entry into the industry in 1985.
- Expanded its client reach from local Ahmedabad operations to establish a verified pan-India commercial distributor network.

Manufacturing Capacity & Localization in India — scored 65/100

SUPPORTING POINTS

- Operates a dedicated production facility at A/4, First Floor, Bileshwar Estate, Part-1, Odhav Road in Ahmedabad, Gujarat.
- Runs a separate commercial assembly and inventory warehouse located at C/2, Balaji Complex, Nehru Park, Vastrapur.

After-Sales Service & Support Network — scored 58/100

SUPPORTING POINTS

- Provides direct multi-channel technical assistance through its centralized Odhav operations office and corporate email channels.
- Manages full-service retail trade verification and support infrastructure via registered vendor accounts on regional trade platforms.

Regulatory Approvals & Certifications — scored 66/100

SUPPORTING POINTS

- Validated under operational Goods and Services Tax registration number 24AAECN9032G1ZL in the state of Gujarat.
- Achieved official vendor trust certification and verification status on major national trade indices including Trade India.

Impact Fire & Safety Appliances Pvt. Ltd. IMPACT ● Evolving Players X 58 Y 47 Overall 52

X — PRODUCT CAPABILITY

Fire-Resistant Battery Chemistry & Materials — scored 25/100

SUPPORTING POINTS

- Company focuses strictly on standalone fire equipment supply/installation rather than manufacturing cell chemistries.

Thermal Runaway Prevention & Suppression Systems — scored 82/100

SUPPORTING POINTS

- Manufactures automatic fire suppression systems including Firetrace, FM200, carbon dioxide, and water mist systems.
- Supplies specialist Class D dry powder extinguishers rated for metal and lithium-based fire containment.

Battery Management System (BMS) Safety Features — scored 30/100

SUPPORTING POINTS

- Offers intelligent addressable fire detection panels and wireless networkable sensors rather than direct cell-level BMS integrations.

Compliance with Indian & International Safety Standards — scored 90/100

SUPPORTING POINTS

- Products are certified to the ISI marking by the Bureau of Indian Standards (BIS).
- Holds international certifications and affiliate memberships from UL (USA), ULC (Canada), CE (Europe), and NFPA (USA).

R&D and Innovation in Battery Fire Safety — scored 65/100

SUPPORTING POINTS

- Actively participated and showcased technical safety portfolios at the Bharat Fire & Safety Congress 2025.

Y — BUSINESS CAPABILITY

Tier-1 OEM Partnerships & Supply Contracts — scored 45/100

SUPPORTING POINTS

- Manufactures electronic valves, tubes, and specialized electronic fire components under its official NIC industry classification code 3210.
- Supplies industrial machinery-grade fire suppression systems and CO2-based fire extinguishers suited for manufacturing environments.

Revenue & Growth in Indian Market — scored 42/100

SUPPORTING POINTS

- Reported an annual operating revenue of ₹2.97 Cr for the financial year 2024, representing a 17.00% year-on-year decline.
- Generated total corporate revenue of less than ₹10 Cr for the financial year ending on March 31, 2025.

Manufacturing Capacity & Localization in India — scored 50/100

SUPPORTING POINTS

- Operates as an unlisted private manufacturing entity incorporated in New Delhi with an authorized share capital of ₹1,00,000.
- Features a fully localized manufacturing and assembly setup categorized under the machinery and equipment sector in India.

After-Sales Service & Support Network — scored 48/100

SUPPORTING POINTS

- Offers direct regional maintenance, specialized fire audits, and safety management programs tailored to the domestic Indian market.
- Deploys field technicians to resolve end-user service challenges regarding compliance and physical appliance upkeep.

Regulatory Approvals & Certifications — scored 52/100

SUPPORTING POINTS

- Maintains complete regulatory filings and corporate status tracking with the Ministry of Corporate Affairs (MCA) in India.
- Standardized its manufacturing outputs under Corporate Identification Number (CIN) U32109DL2004PTC127259 to comply with Indian industrial safety codes.

Parameter Definitions

X — Product Capability

1. **Fire-Resistant Battery Chemistry & Materials**
   Use of inherently safer battery chemistries (e.g., LFP, solid-state) and flame-retardant materials in cell and pack design to reduce fire risk. Evidence includes material specifications, safety test results (e.g., nail penetration, overcharge), and third-party certifications.
2. **Thermal Runaway Prevention & Suppression Systems**
   Design features and active/passive systems that prevent or mitigate thermal runaway, such as cooling systems, venting, and fire suppression agents. Strength shown by successful test outcomes (e.g., no fire propagation) and integration in commercial products.
3. **Battery Management System (BMS) Safety Features**
   Advanced BMS with real-time monitoring, fault detection, and fail-safe mechanisms to prevent overcharging, overheating, and short circuits. Evidence includes BMS algorithms, sensor accuracy, and response times in safety-critical scenarios.
4. **Compliance with Indian & International Safety Standards**
   Adherence to standards like AIS-156 (India), IEC 62619, UL 1973, and UN38.3 for battery safety. Strength indicated by certifications, test reports, and compliance with evolving Indian regulations.
5. **R&D and Innovation in Battery Fire Safety**
   Ongoing research and development focused on improving battery safety, including new materials, designs, and testing methodologies. Evidence includes patents, R&D expenditure, collaborations with research institutions, and published innovations.

Y — Business Capability

1. **Tier-1 OEM Partnerships & Supply Contracts**
   Contracts and partnerships with major Indian electric vehicle (EV) manufacturers, energy storage integrators, or electronics OEMs. Evidence includes named customers, contract values, and duration of agreements.
2. **Revenue & Growth in Indian Market**
   Sales revenue and growth rate specifically from the Indian market for battery products with fire safety features. Evidence includes financial statements, investor presentations, and market share reports.
3. **Manufacturing Capacity & Localization in India**
   Production capacity and degree of local manufacturing in India, including cell and pack assembly. Evidence includes factory locations, capacity figures (GWh), and localization commitments under PLI schemes.
4. **After-Sales Service & Support Network**
   Extent of service centers, technical support, and warranty offerings across India. Evidence includes number of service points, response times, and customer satisfaction metrics.
5. **Regulatory Approvals & Certifications**
   Obtained approvals and certifications from Indian regulatory bodies (e.g., ARAI, ICAT) and international organizations. Evidence includes certificate numbers, approval dates, and compliance with BIS standards.

Research Methodology

This quadrant was compiled independently through structured desk research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1 Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

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2 Desk and Primary Research

Desk research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

→

3 Screening

Each candidate was screened against the market's inclusion criteria and required commercial role, and its market activity was verified. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicate entries were consolidated.

→

4 Evidence Compilation

Every screened company was assessed on all ten parameters. For companies profiled on the chart, evidence was recorded parameter by parameter. Findings from B2B marketplace listings are reported as primary research; items not verified through primary research are marked as such.

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5 Scoring

Each parameter is rated on a 0–100 scale against its written definition, based on the evidence available for that company. A company's X and Y scores are the simple average of its five parameter ratings on each axis, and its overall score is the average of X and Y.

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6 Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.